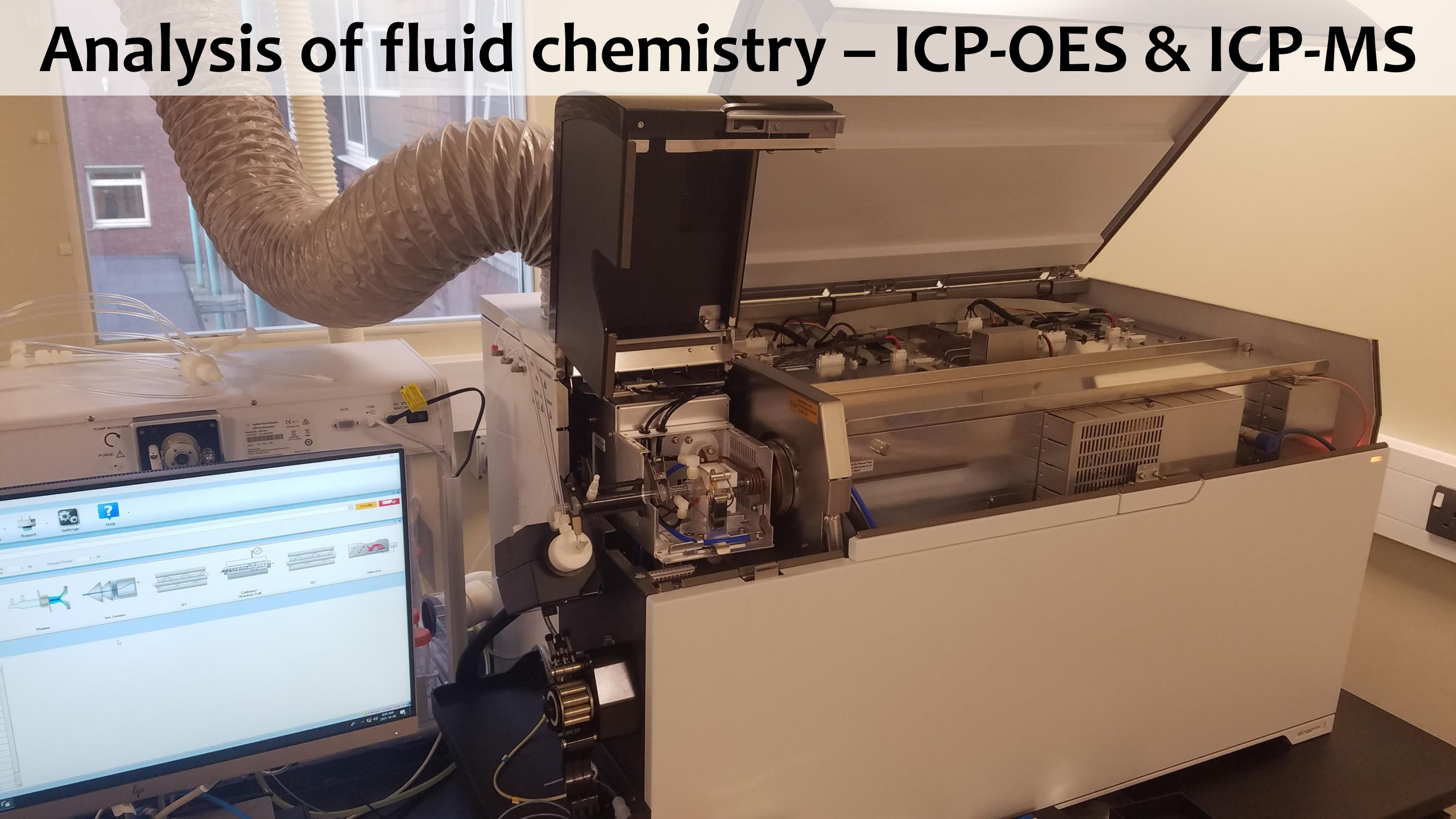
Analysis of fluid chemistry – ICP-OES & ICP-MS



What is ICP-OES?

Inductively **C**oupled **P**lasma – **O**ptical **E**mission **S**pectroscopy (sometimes **A**tomic **E**mission **S**pectroscopy – AES)

<u>Why using ICP-OES</u>?

1) Almost no bonds survive plasma heat, no chemical interferences (*well…*).
2) Background is low, so low detection limits.
3) Simple linear calibration over several orders of magnitude.

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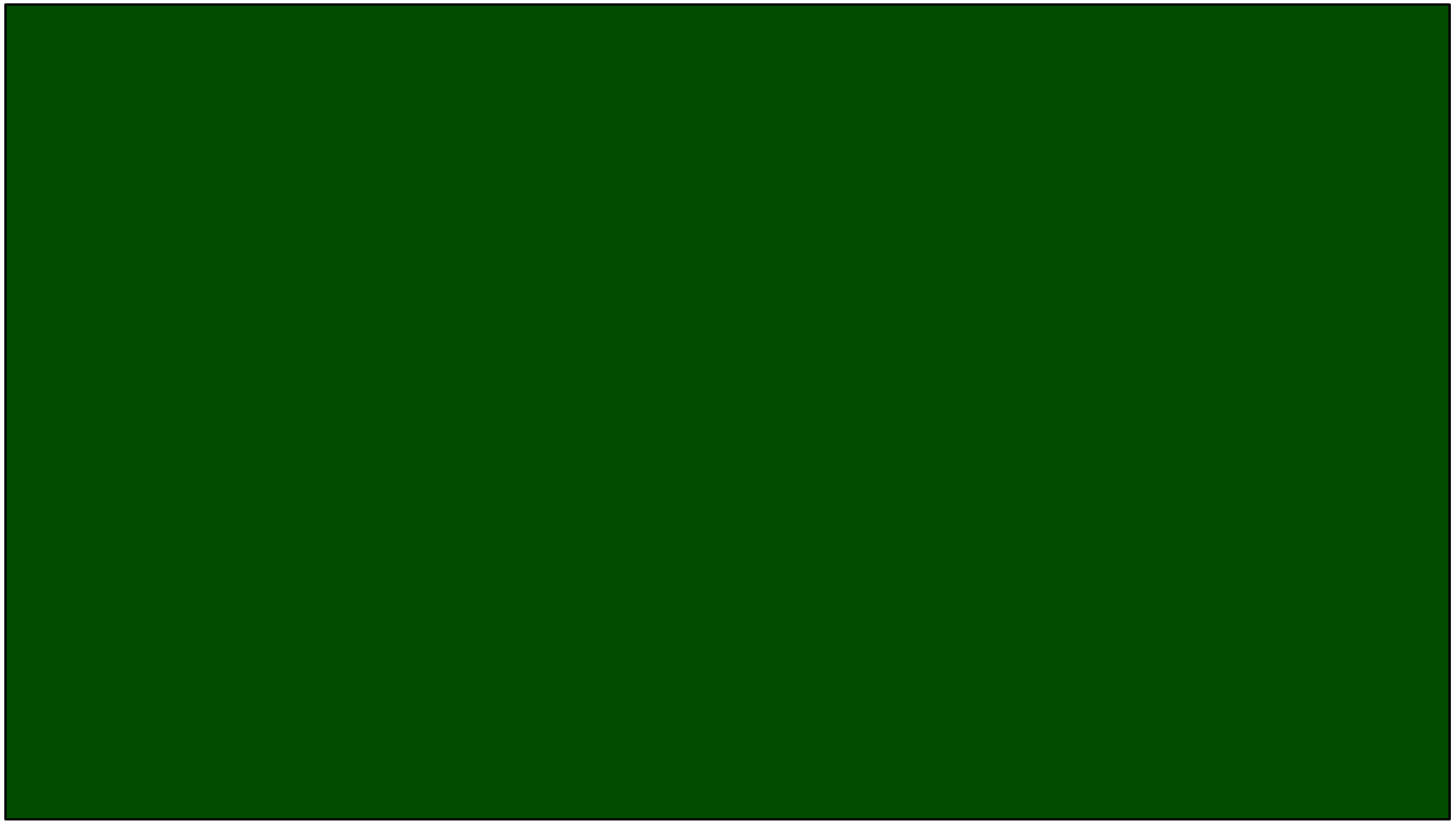
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Key applications

- Environmental problems involving aqueous phase.
- Geochemical problems requiring large data set.
- Experimental fluid analysis.

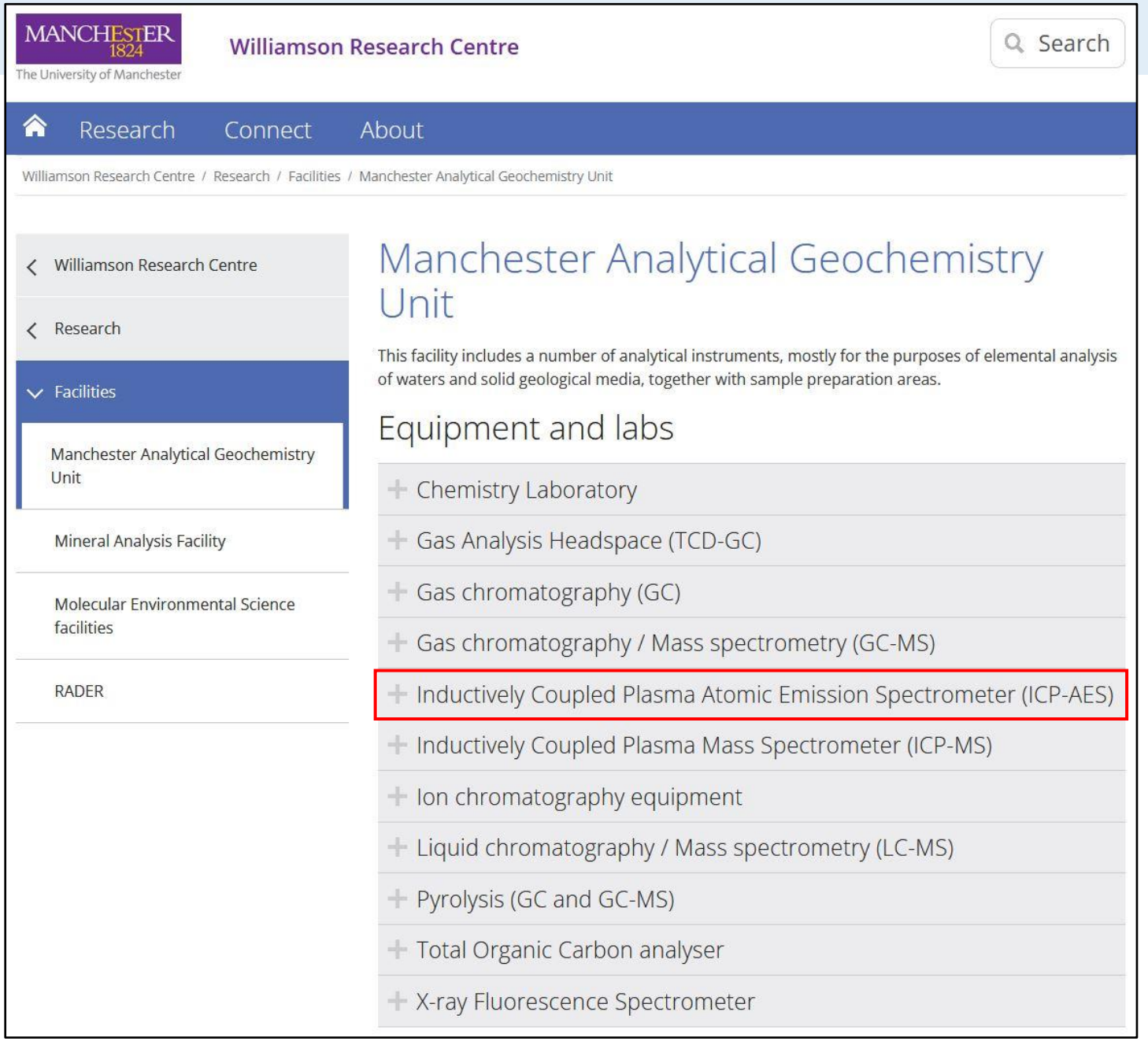
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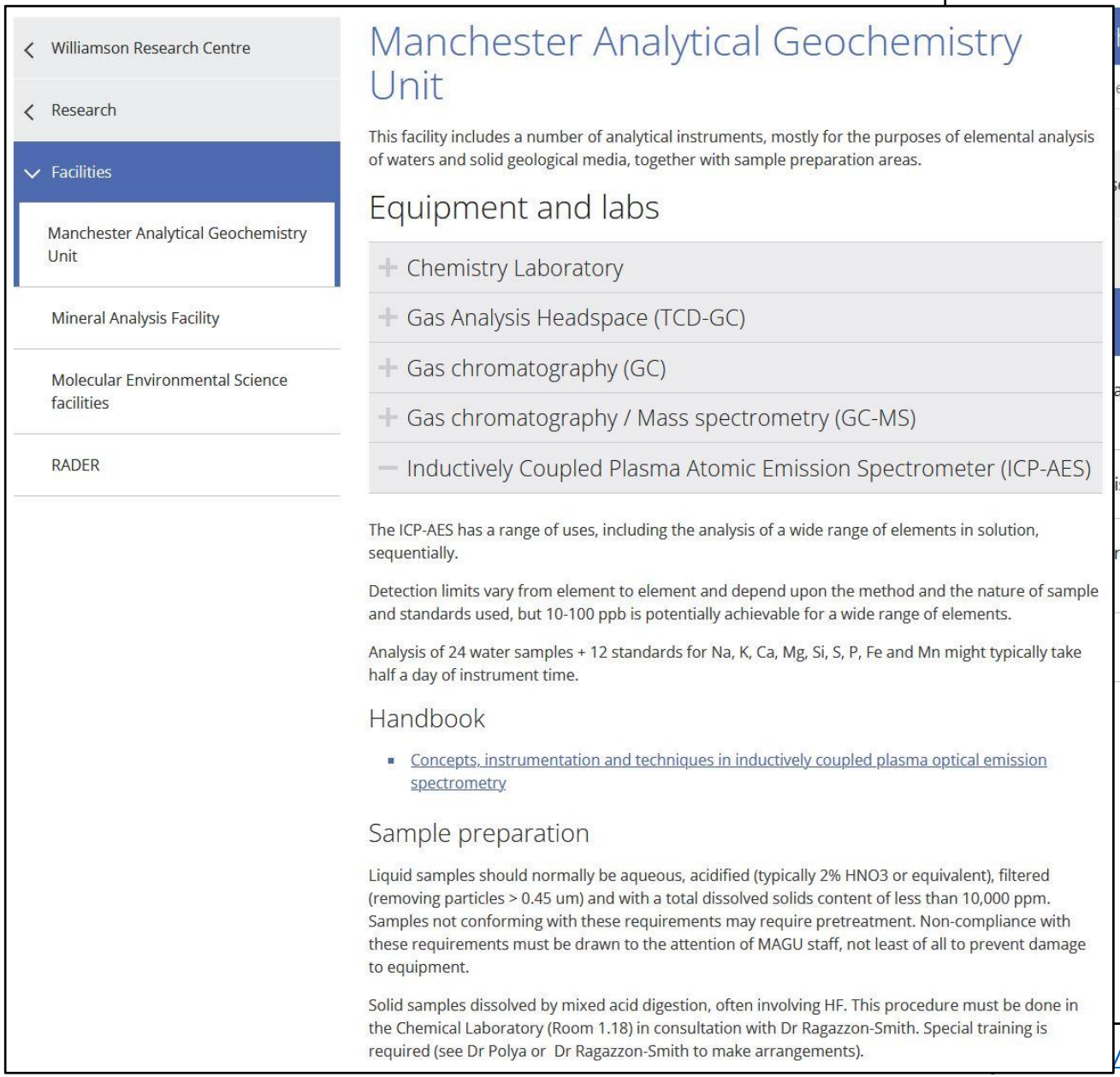
https://www.spectro.com/icp-oes-principle

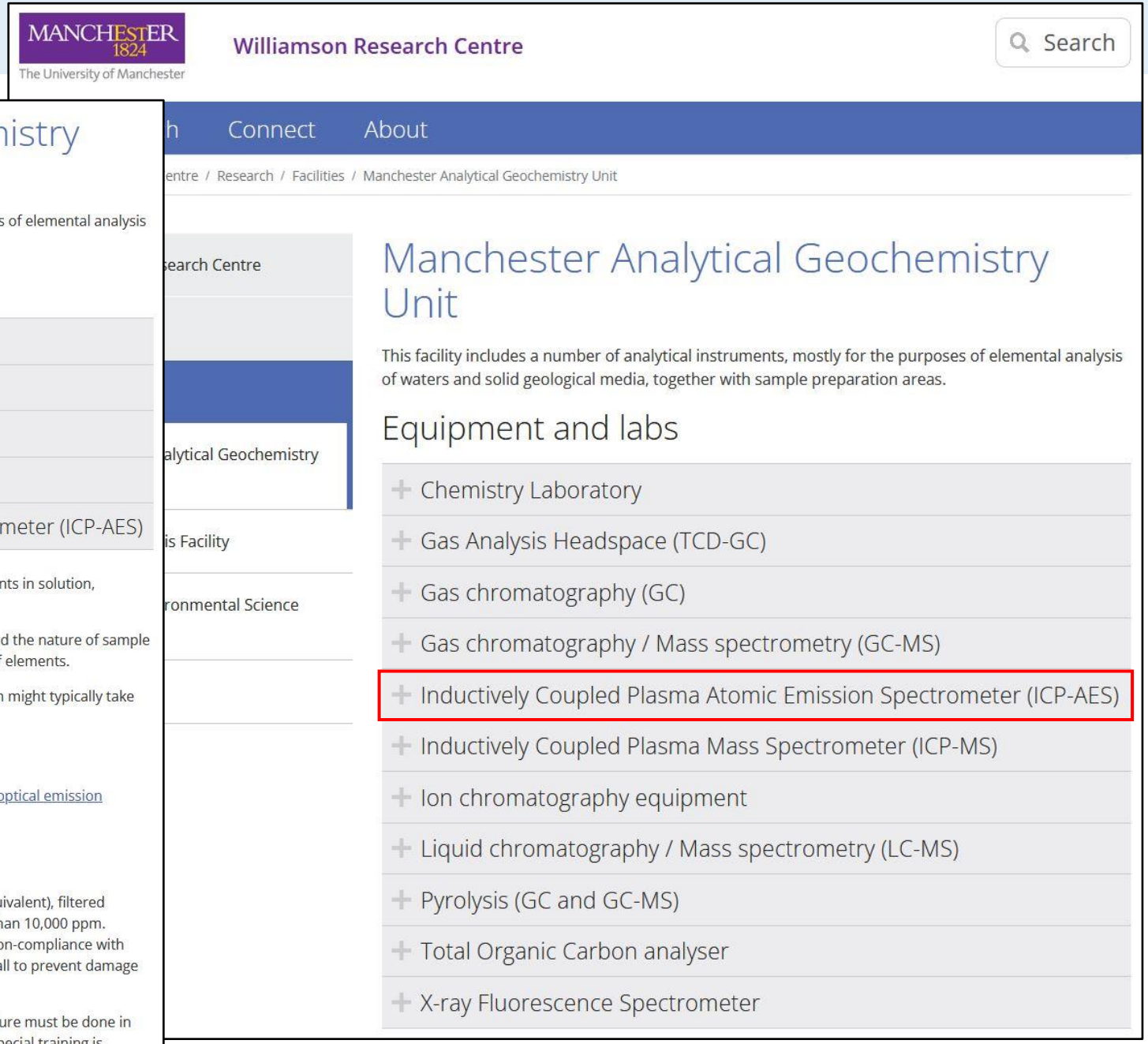
ICP-OES in Manchester



https://www.ees.manchester.ac.uk/wrc/research/facilities/magu/

ICP-OES in Manchester





w.ees.manchester.ac.uk/wrc/research/facilities/magu/

ICP-OES in Manchester

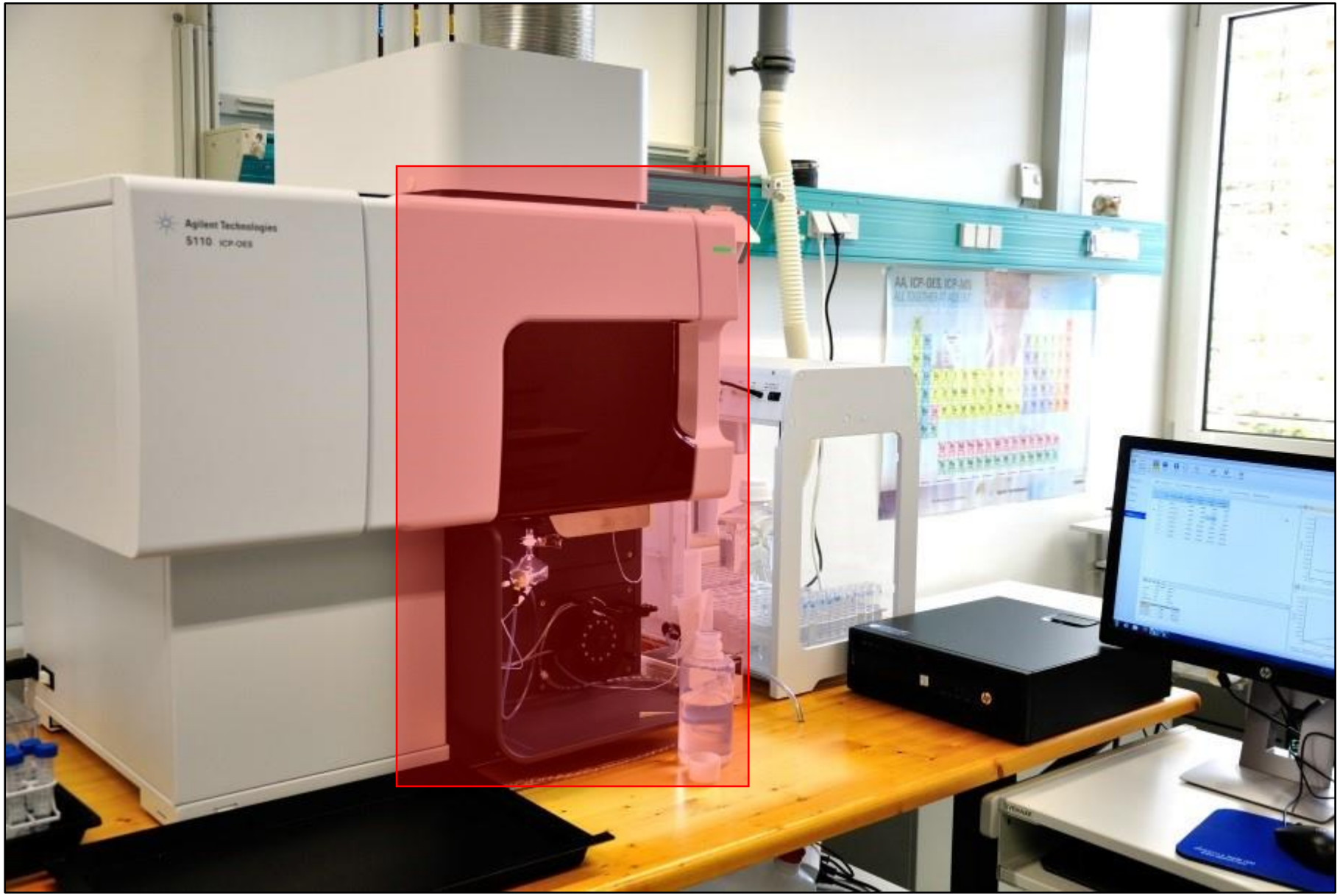


Key components:

1) Source with plasma torch (energy input)
2) Spectrometer (separates emission lines and measures intensities)
3) Computer (record data and convert intensity to concentration)

https://www.ees.manchester.ac.uk/wrc/research/facilities/magu/

ICP-OES in Manchester

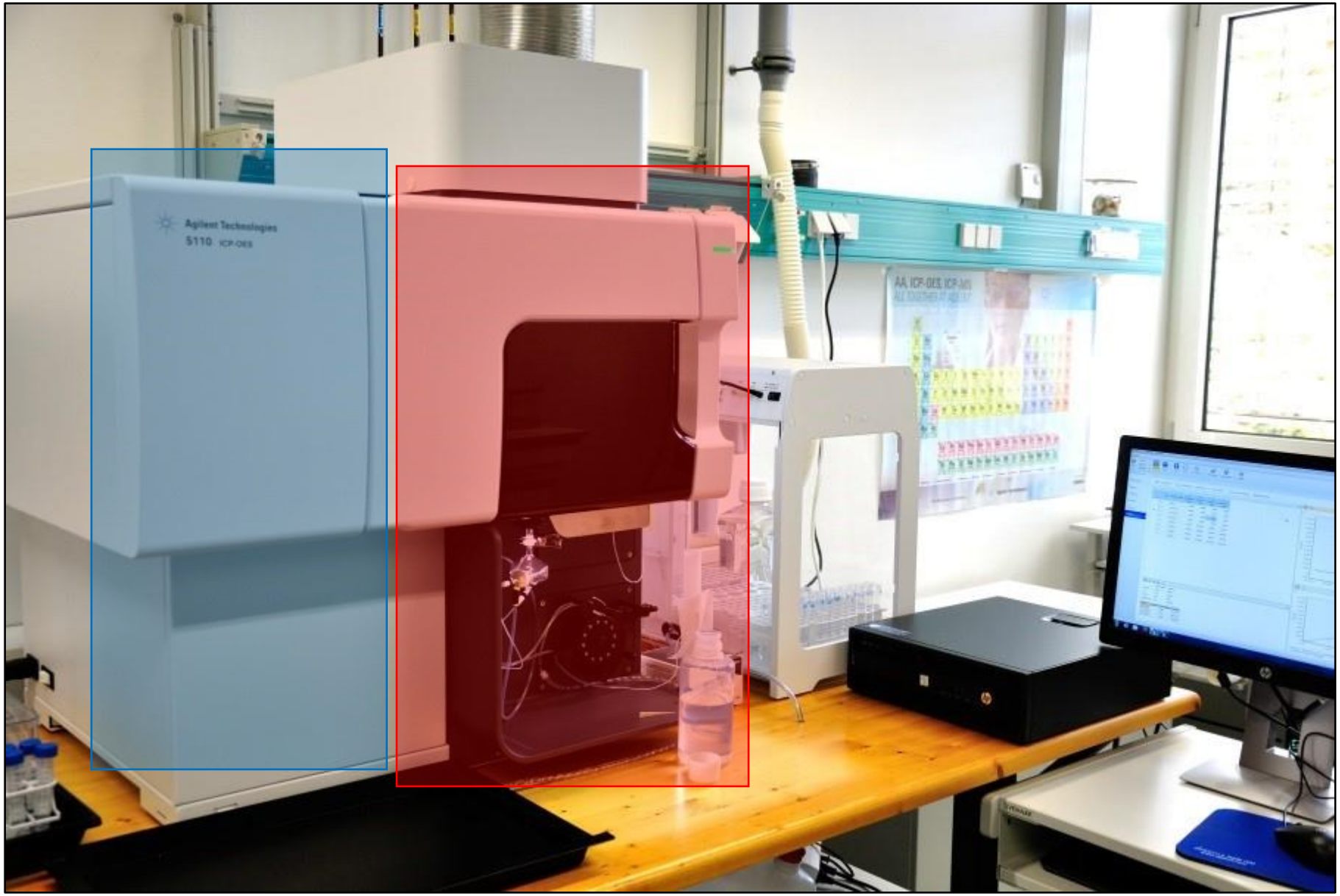


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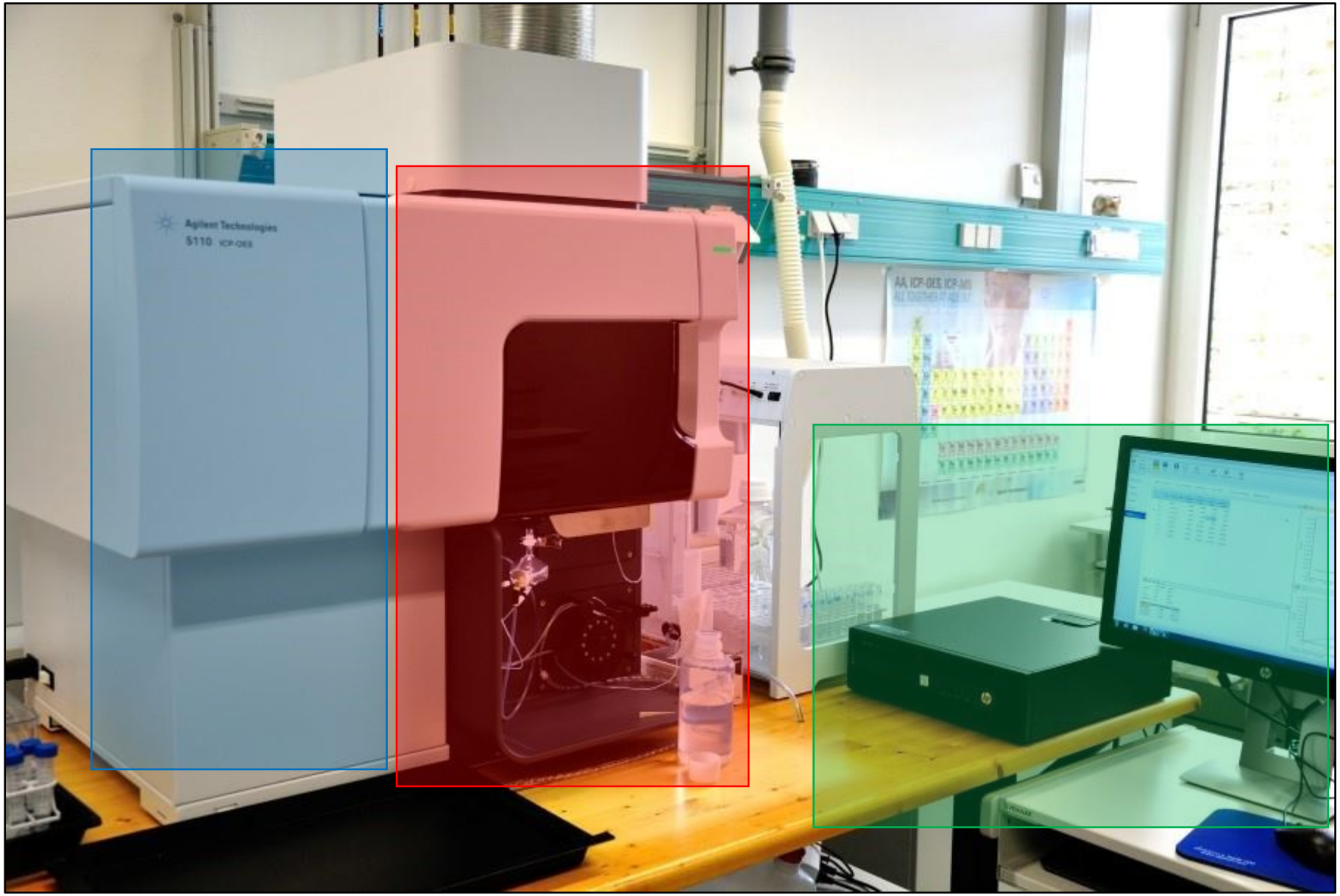


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ICP-OES – schematics

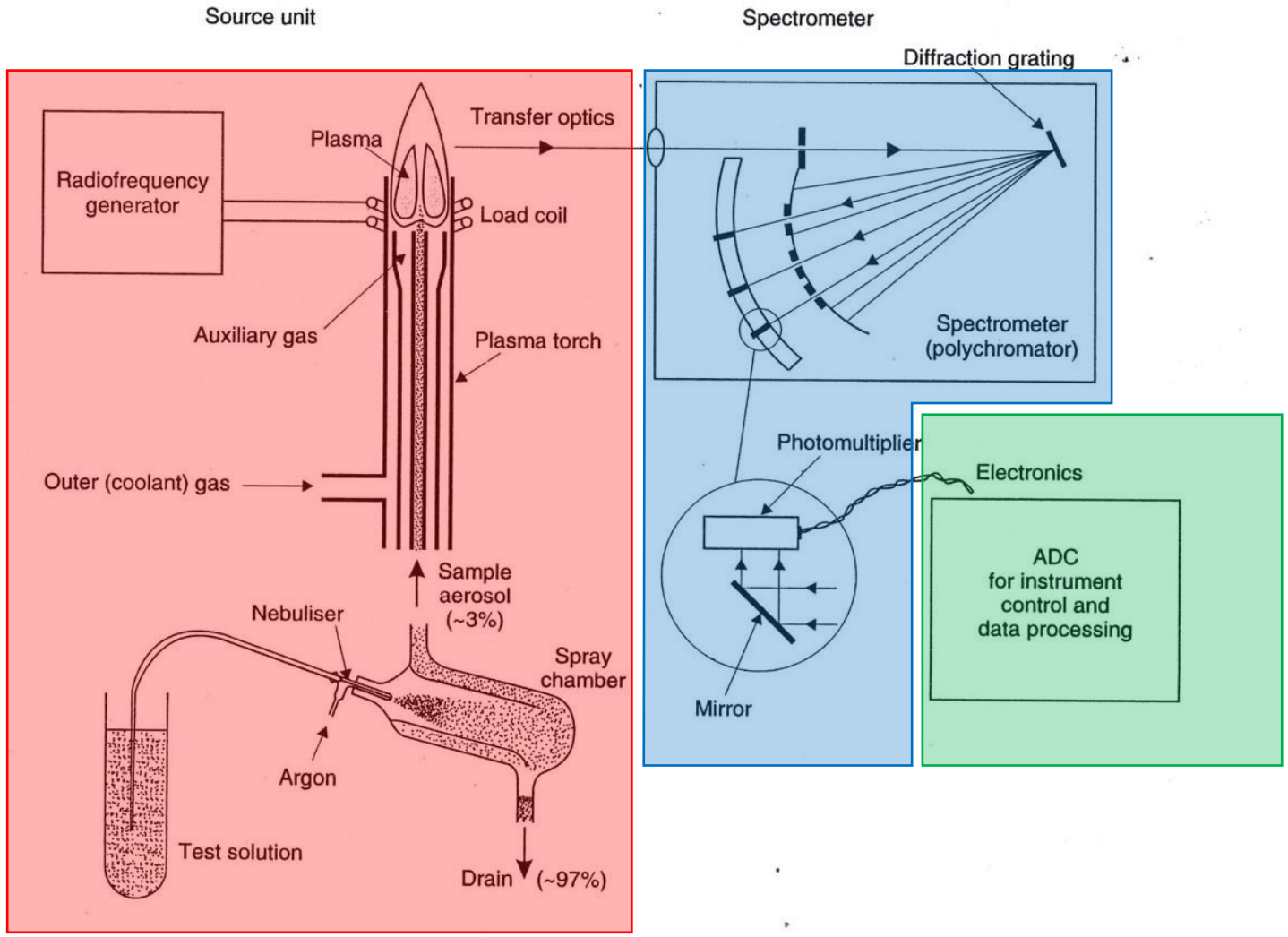
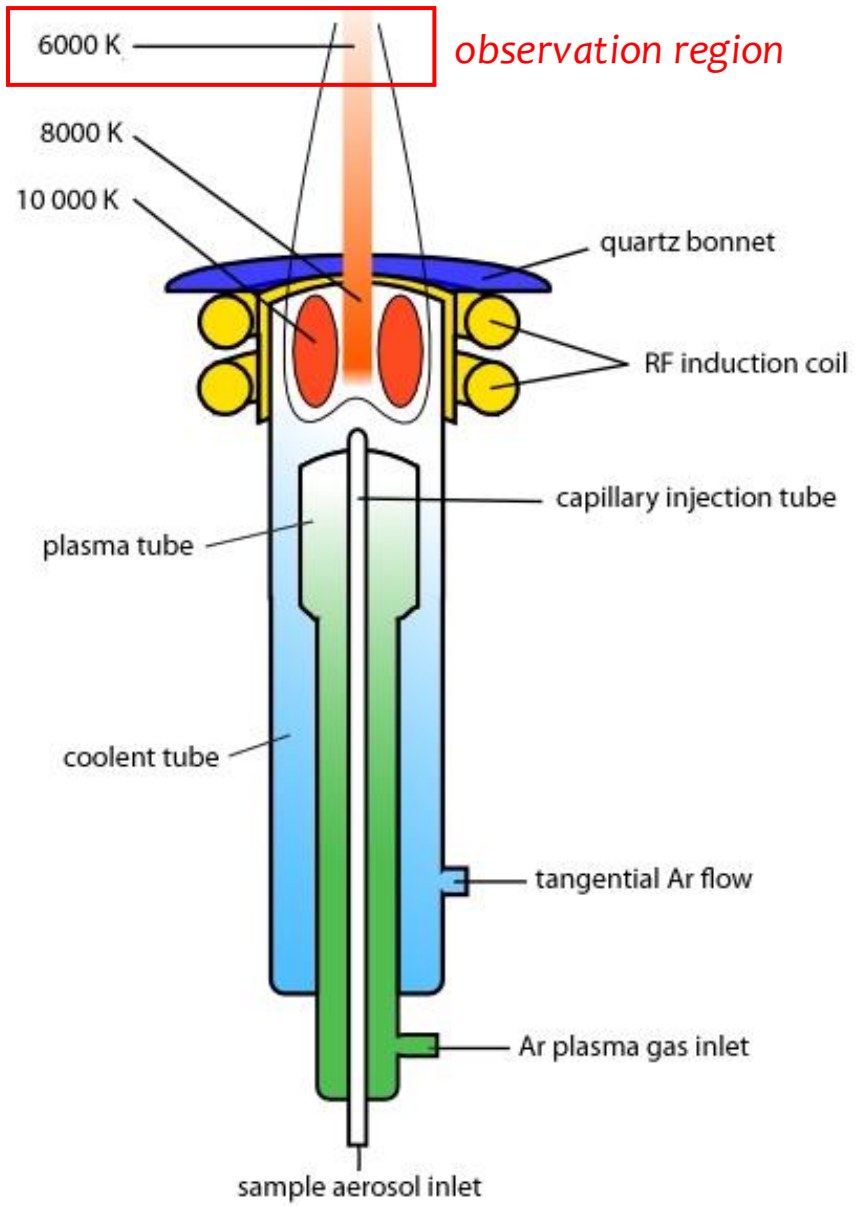


Figure 4.1 Schematic diagram showing the major components of a typical multi-element ICP–AES system. The analyte solution is introduced into the plasma as an aerosol suspended in argon gas. Emitted light from the elements present in the plasma is resolved into its component wavelengths by a multi-channel optical spectrometer. **Photomultiplier** tubes or **charge-coupled devices** (CCDs) measure the light intensity at each designated wavelength, and comparison with standard samples allows element concentrations to be measured. Reprinted with kind permission of Chapman & Hall from *Handbook of Inductively Coupled Plasma Spectroscopy*, M. Thompson and J. N. Walsh, 1989, Blackie and Son Ltd.

Key components:

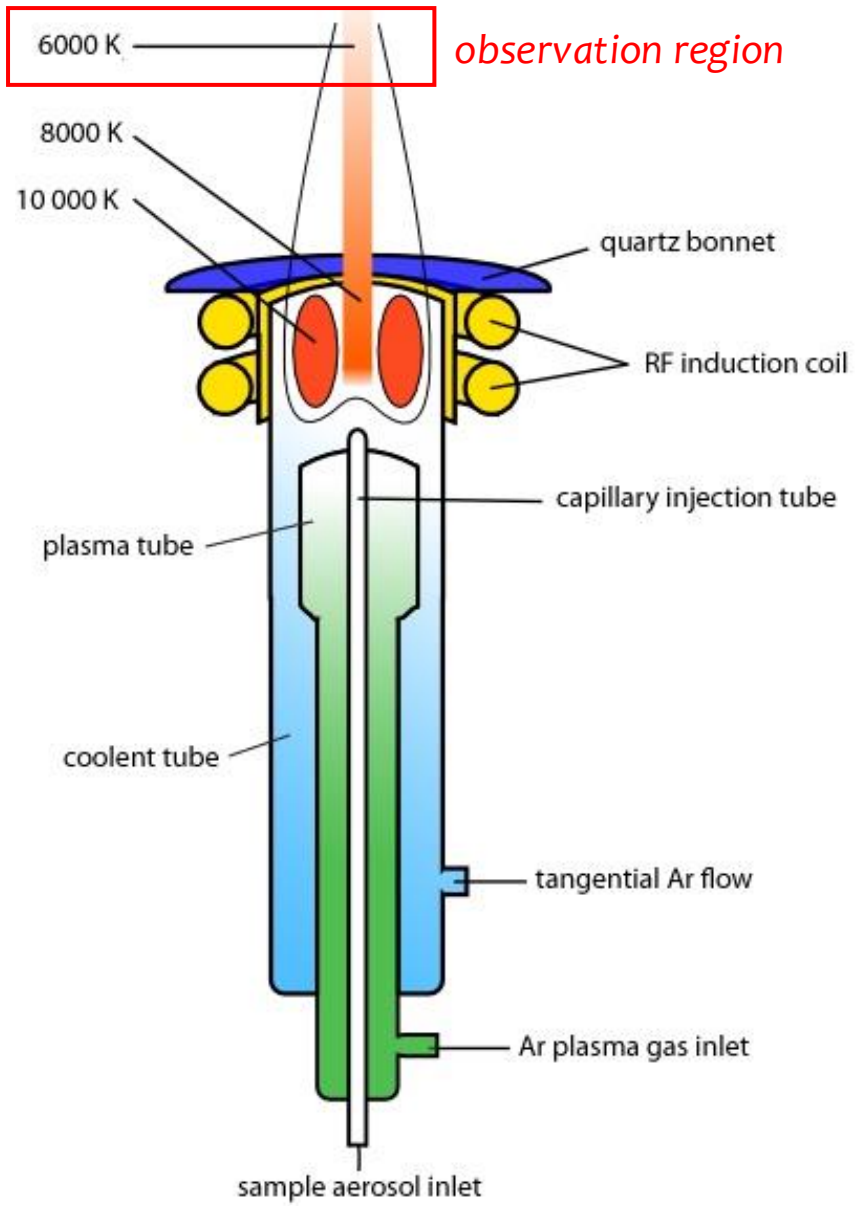
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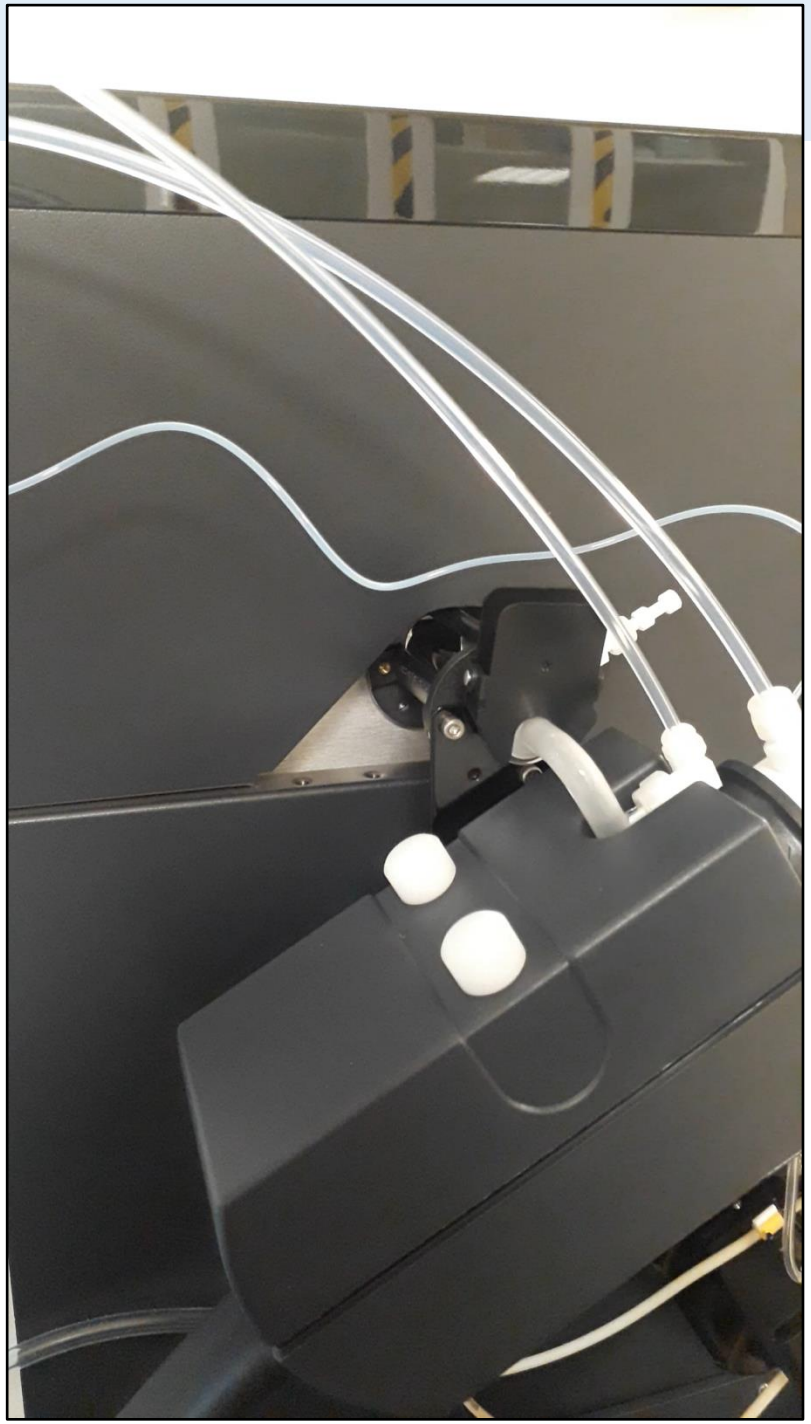
ICP-OES – torch and plasma



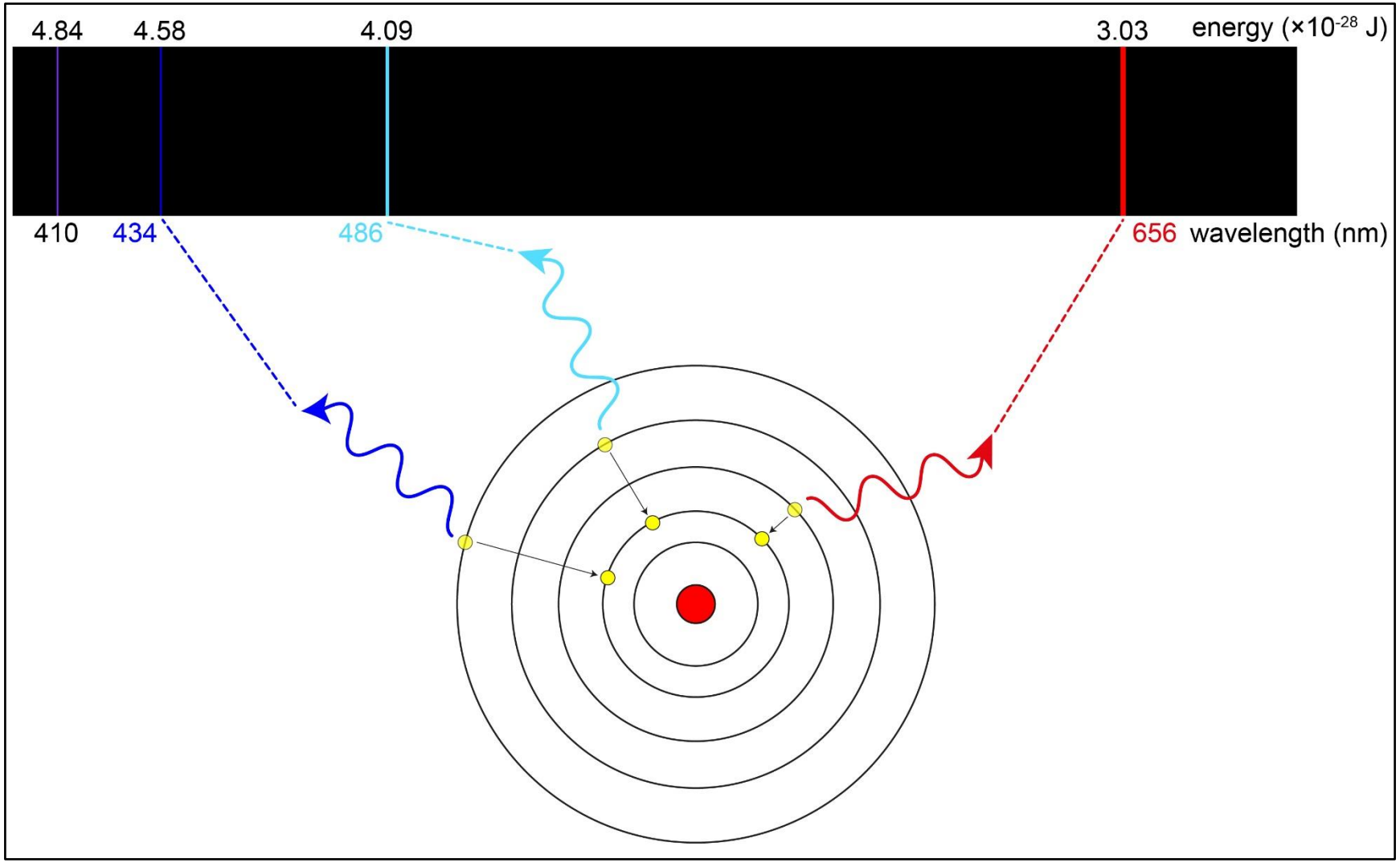
- Argon (Ar) gas flows in torch.
- Radio-frequency (RF) power applied to coil creates electric and magnetic fields at the end of the torch.
- Spark **strips electrons from Ar gas**; these electrons accelerate and increase energy in magnetic field; they **collide with Ar atoms** and strip off more electrons; eventually plasma contains Ar atoms, Ar^+ ions, and electrons.
- Introduced aerosol is (1) ***desolvated*** in solid molecules, which are then (2) ***vaporised*** in gas molecules, then (3) ***atomised*** into atoms.
- Finally, atoms are ***excited*** and/or ionised.

ICP-OES – torch and plasma



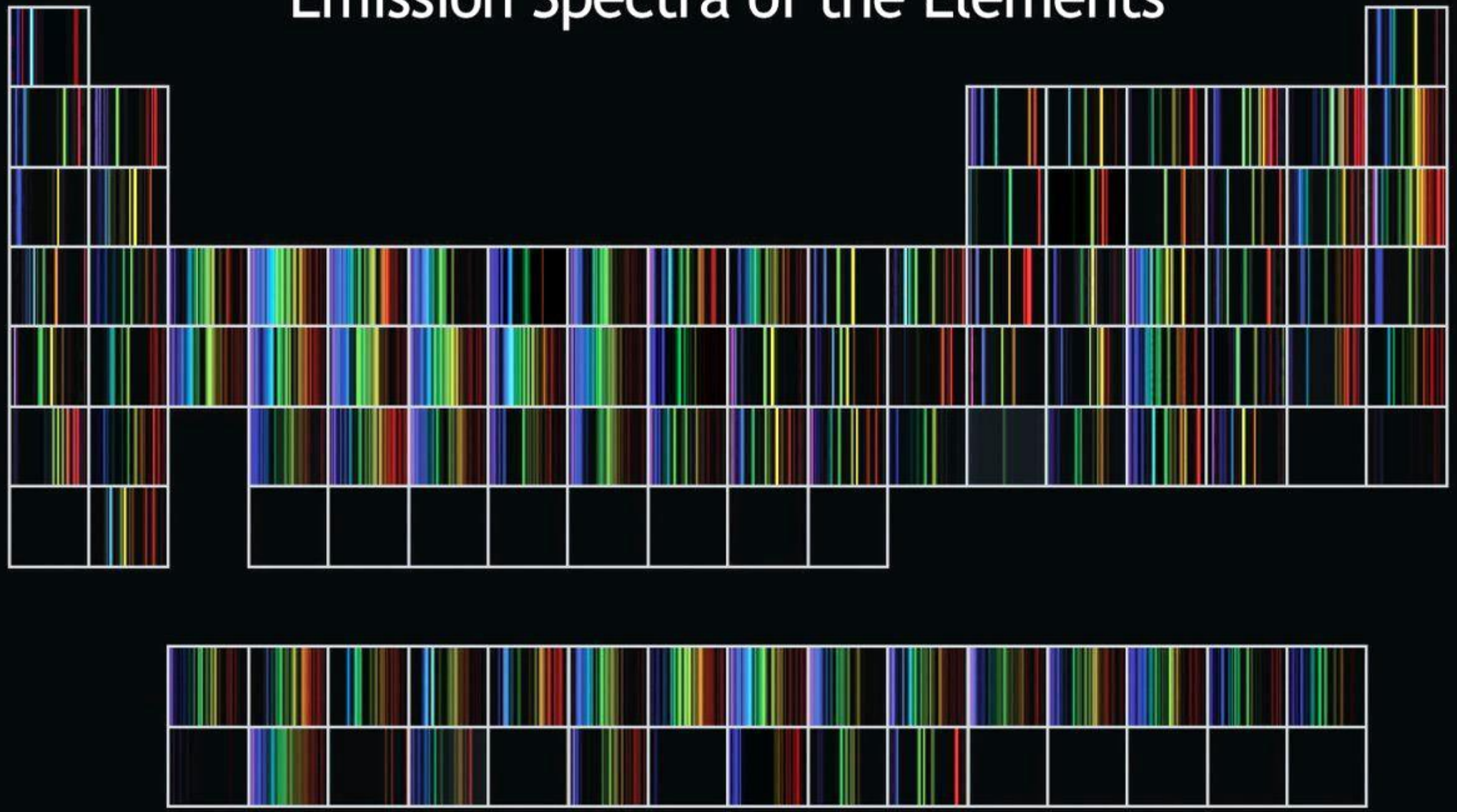


ICP-OES – photon emission: example of Hydrogen



When electrons in excited atoms relax back to ground state, they give up energy as photons with **characteristic wavelengths**.

Emission Spectra of the Elements

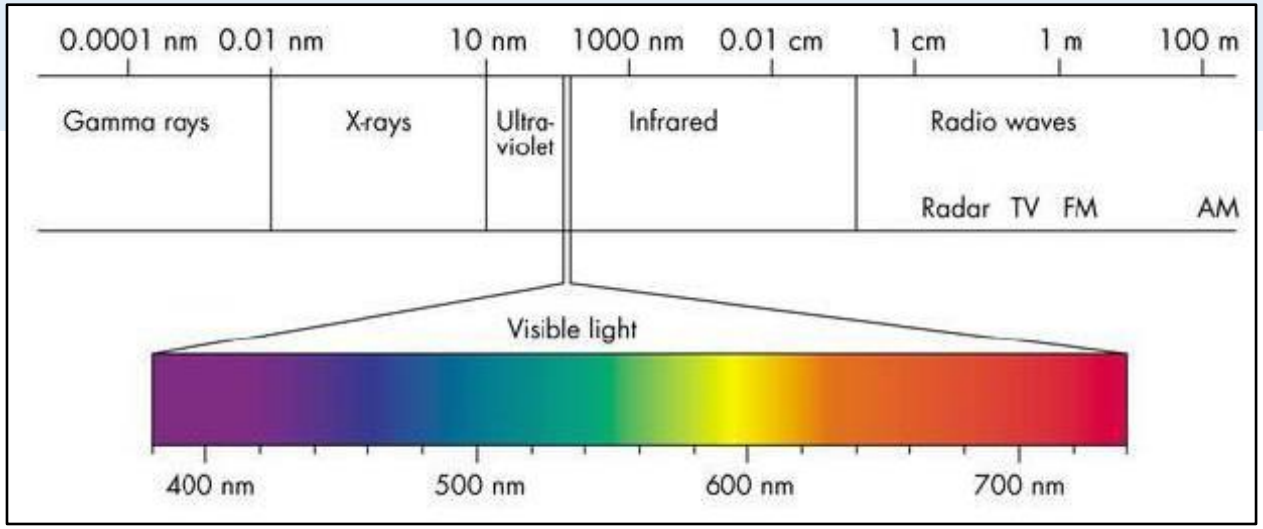


No two elements will emit light at the same wavelengths.

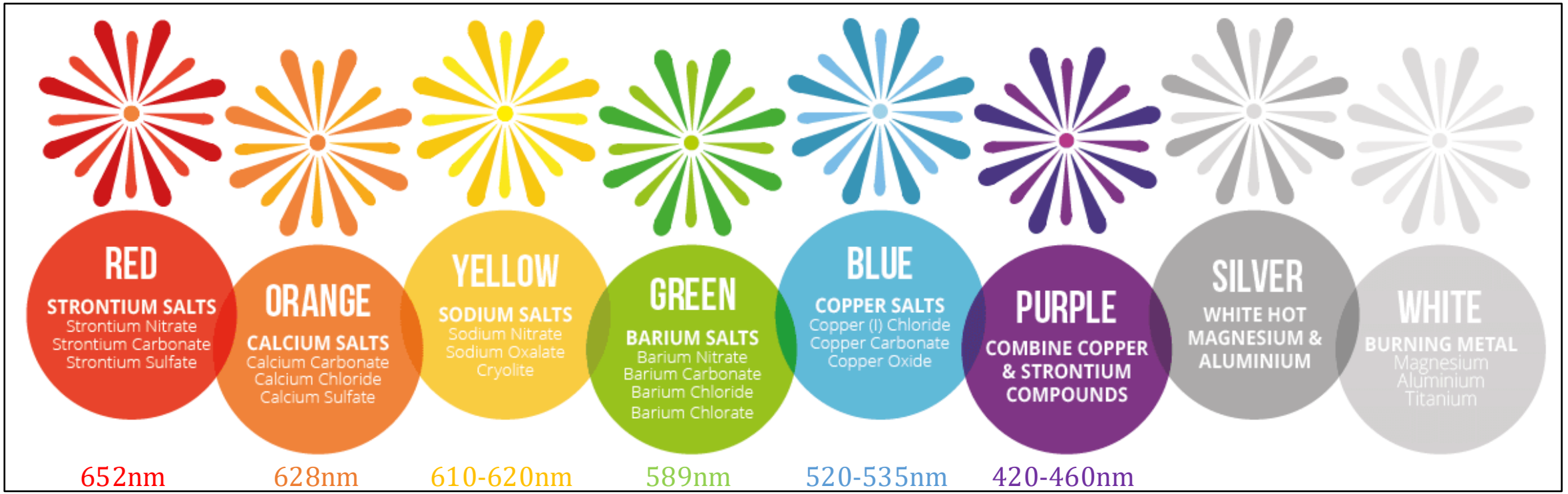
field tested systems
REAL-TIME CLASSROOM SPECTROSCOPY

ICP-OES – photon emission

Colours you see during fireworks are produced by elements with characteristic emissions in visible light as listed below.



https://munsell.com/color-blog/chemistry-fireworks-colors/



https://earthsky.org/human-world/how-do-fireworks-get-their-vibrant-colors/

ICP-OES – useful energy equations

$$\mathbf{E = \hbar\upsilon}$$

Energy = Planck's constant × Frequency
with $\hbar = 6.626 \times 10^{-34}$ J.s

$$\boldsymbol{\lambda\upsilon = c}$$

Wavelength × Frequency = speed of light
with $c = 3 \times 10^{8}$ m s^{-1} and 1 eV = 1.6022×10^{-19} J

$\mathbf{E = \hbar c / \lambda}$ → 12.4 keV @ 1 Å and 12.4 eV @ 100 nm

ICP-OES – spectrometer detection systems

- Slits and mirrors focus light emitted at torch into well-defined beam.
- Grating disperses polychromatic light beam according to wavelength.

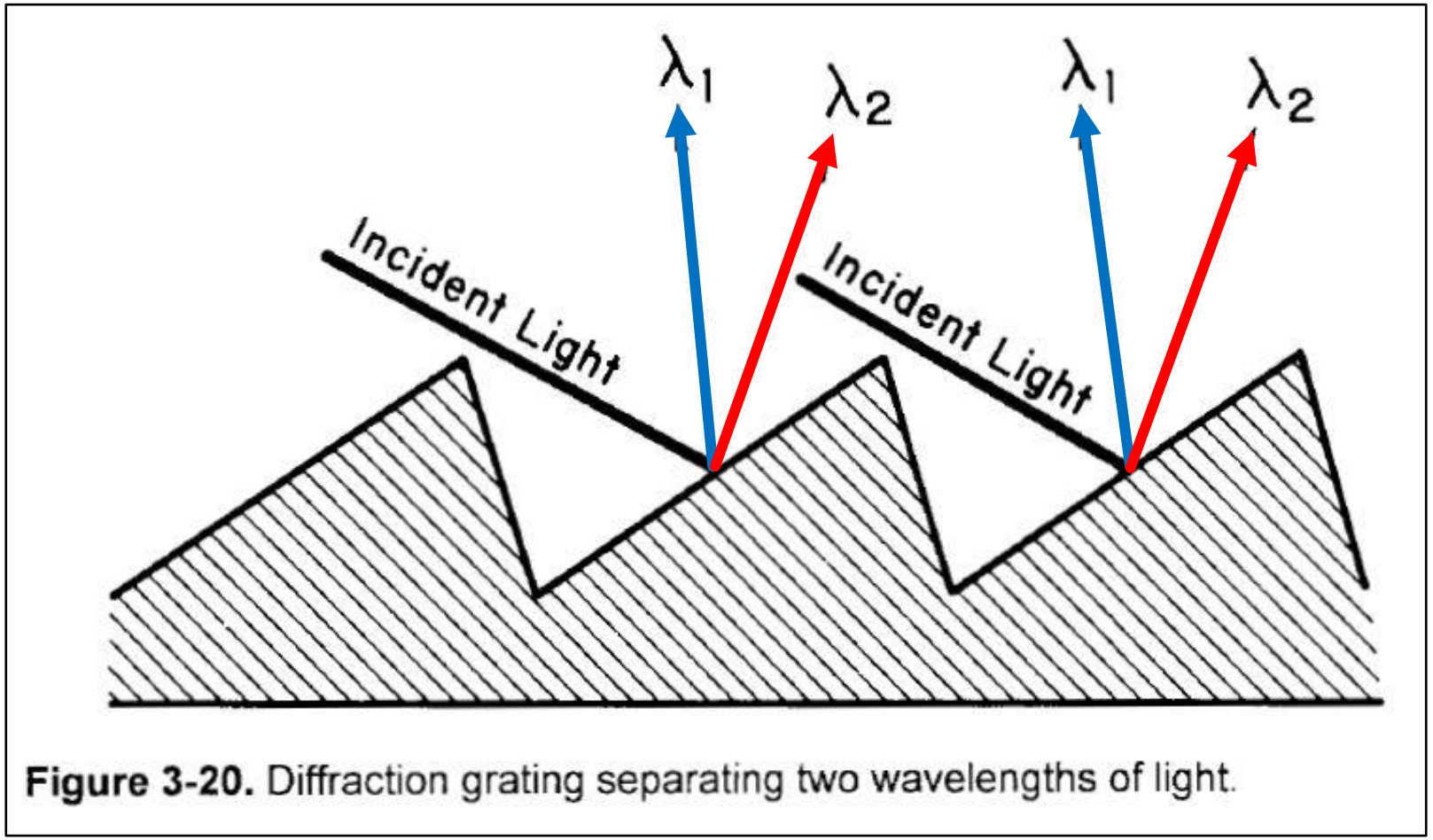


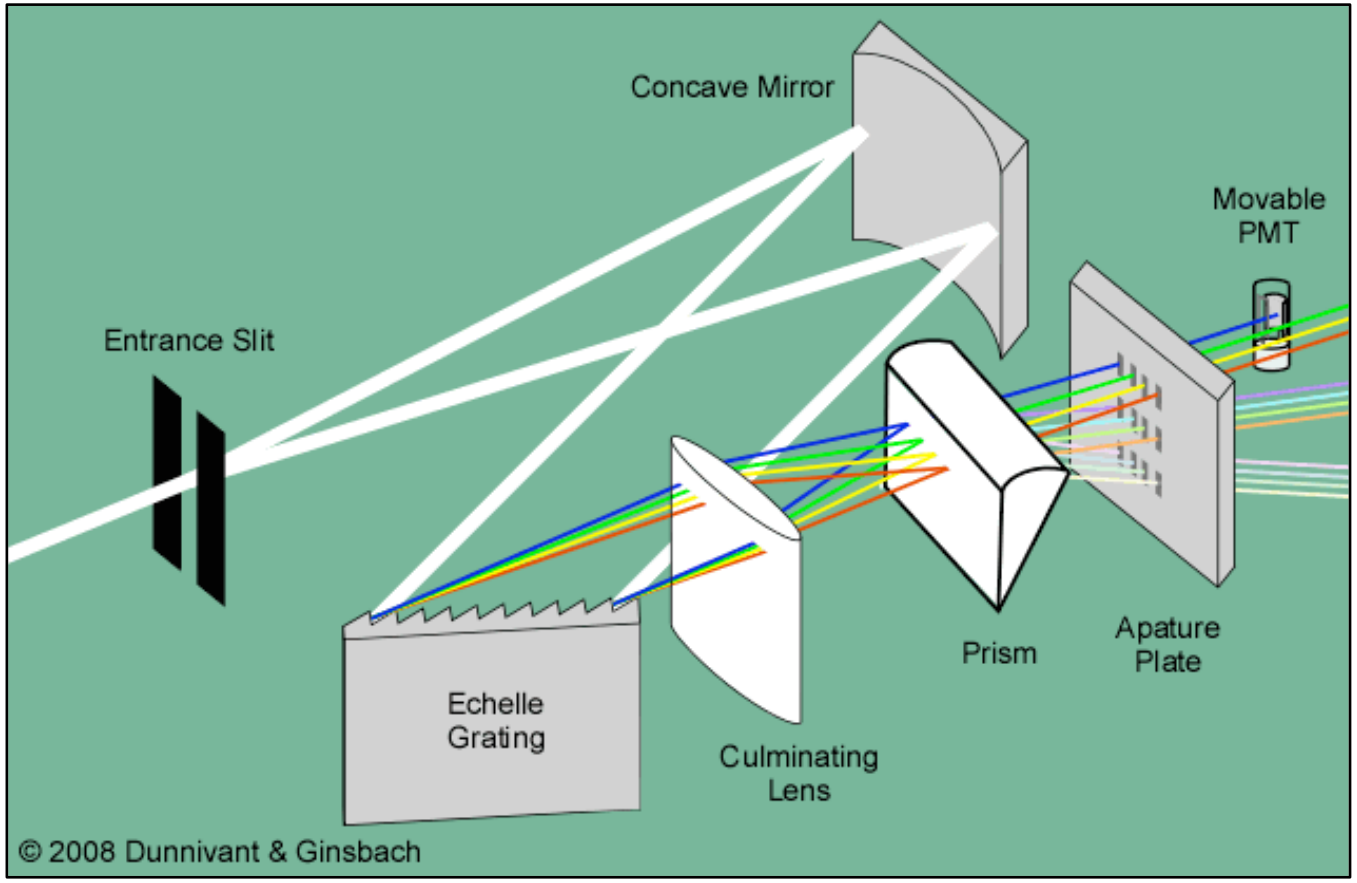
Figure 3-20. Diffraction grating separating two wavelengths of light.

Boss & Fredeen (2004) Concepts, Instrumentation and Techniques in Inductively Coupled Plasma Optical Emission Spectrometry.

ICP-OES – spectrometer detection systems

Two main types:

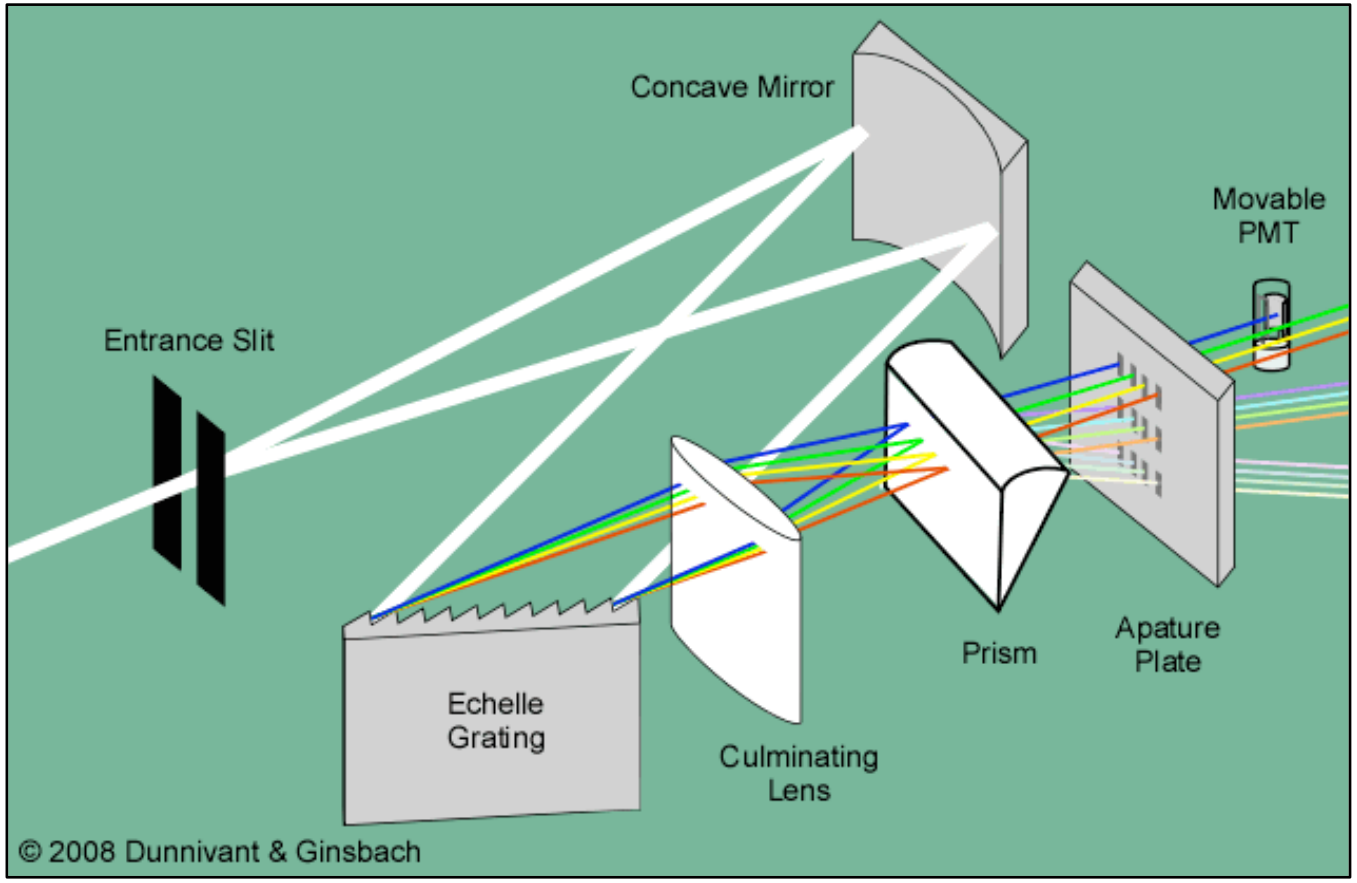
- Monochromators: 1 element at a time, grating rotates for multi-elements.

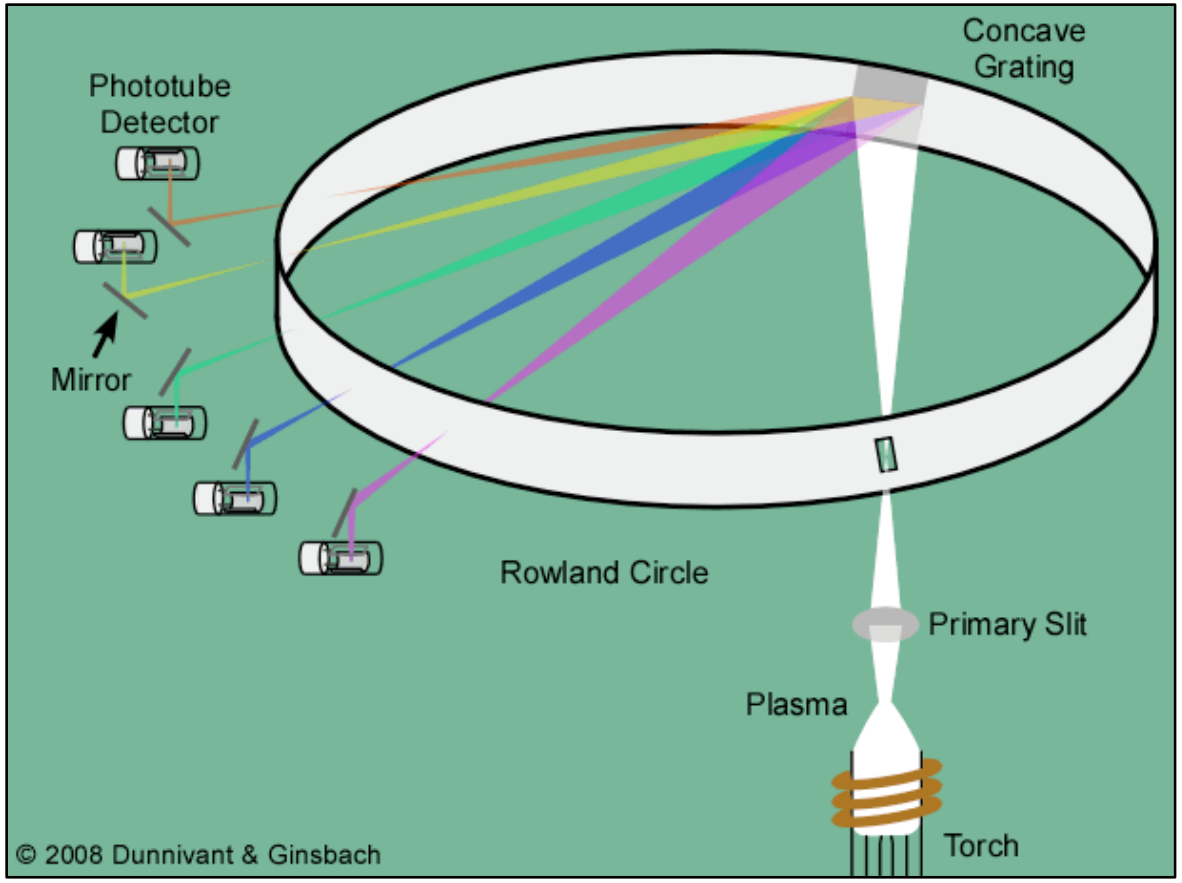


ICP-OES – spectrometer detection systems

Two main types:

- Monochromators: 1 element at a time, grating rotates for multi-elements.
- Polychromators: several elements simultaneously (e.g., Rowland circle).





ICP-OES – background and spectral interferences

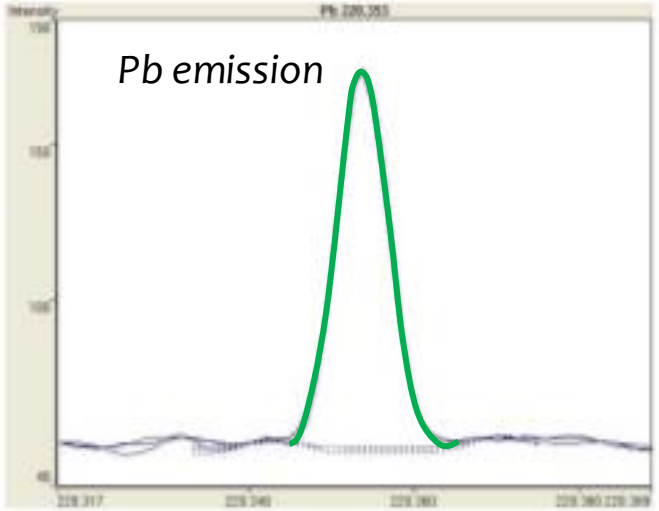
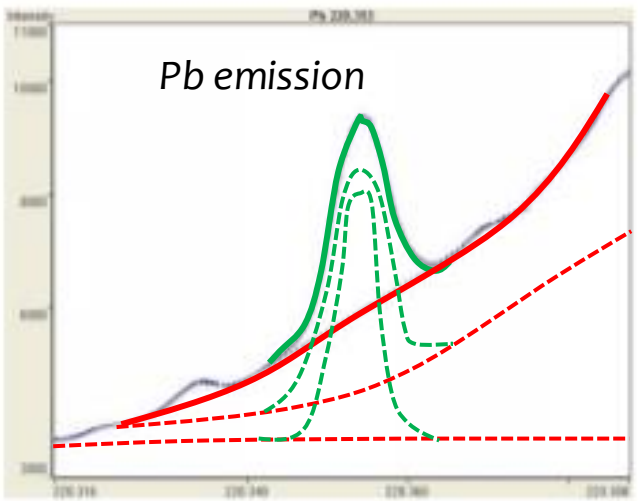
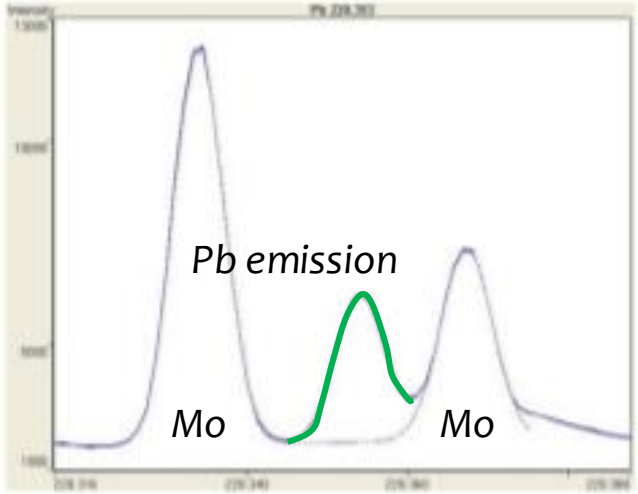


Figure 2a. Pb 220.353 nm emission line in deionized water. Simple background spectrum with either OPBC or FBC being suitable.



High Al content in matrix shifts background

Figure 2b. Pb 220.353 nm emission line in 2000 mg/L aluminum. Wing overlap from spectral-line broadening of Al 220.467 nm has resulted in a higher background signal with a curved profile at Pb 220.353 nm, which FBC successfully corrects for.



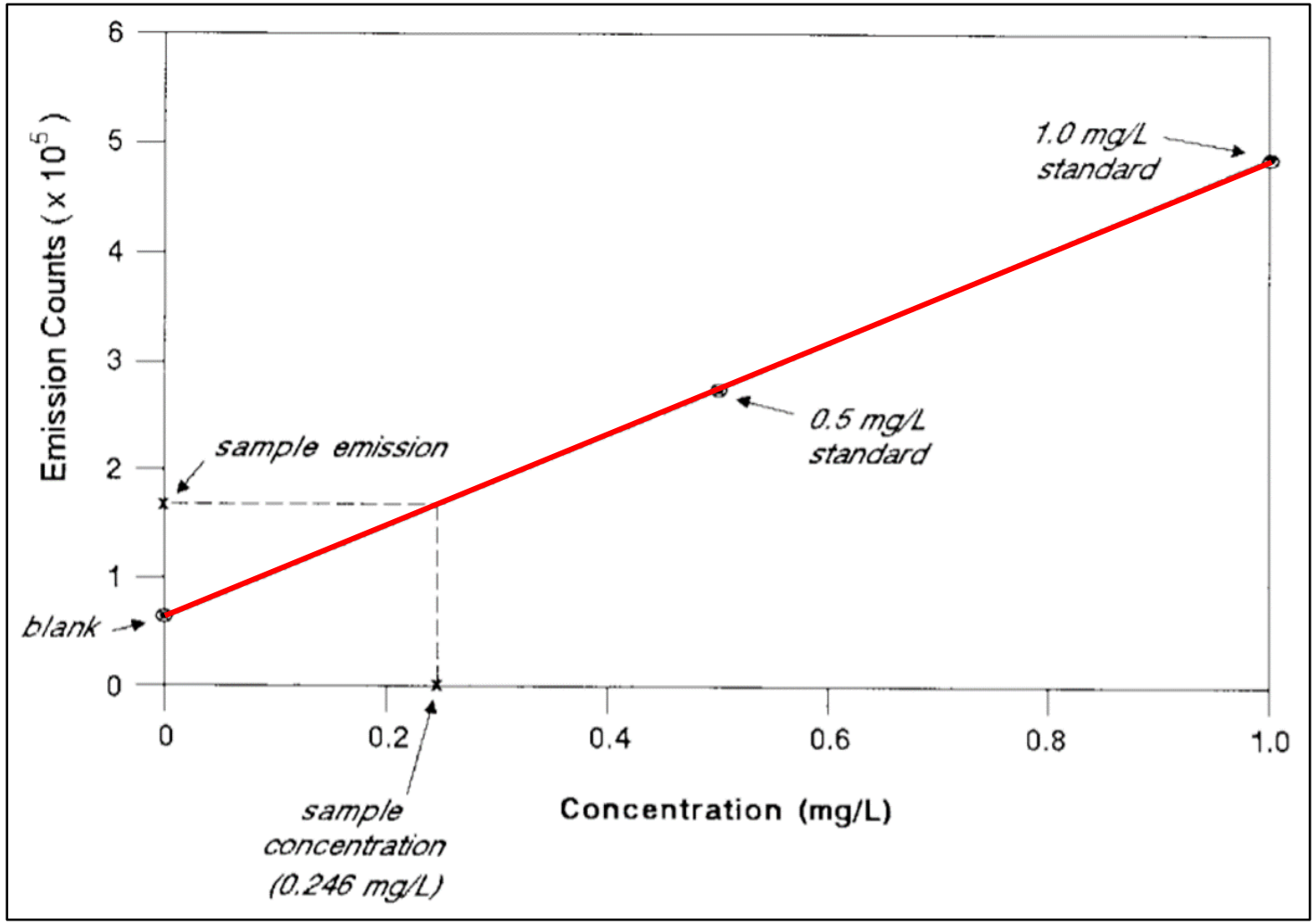
Mo spectral overlap, shifts background

Figure 2c. Pb 220.353 nm emission line in 1000 mg/L molybdenum. Nearby interfering emission lines of Mo make using OPBC near impossible. Again, this is no problem for Agilent's fitted background correction.

- Composition of the matrix fluid very important, both in terms of consistency between standards and samples, and potential presence of unwanted species that can interfere with elements of interest.

ICP-OES – abundance quantification

- Linear dynamic range 4-6 orders of magnitude, **if** matrix is constant.



Boss & Fredeen (2004) Concepts, Instrumentation and Techniques in Inductively Coupled Plasma Optical Emission Spectrometry.

Elements measurable by ICP-OES

Detection limit range

- <0.1 ppb (µg/L)
- 0.1-1 ppb (µg/L)
- 1-10 ppb (µg/L)

hydrogen 1 H 1.0079																	helium 2 He 4.0026
lithium 3 Li 6.941	beryllium 4 Be 9.0122											boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180
sodium 11 Na 22.990	magnesium 12 Mg 24.305											aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948
potassium 19 K 39.098	calcium 20 Ca 40.078	scandium 21 Sc 44.956	titanium 22 Ti 47.867	vanadium 23 V 50.942	chromium 24 Cr 51.996	manganese 25 Mn 54.938	iron 26 Fe 55.845	cobalt 27 Co 58.933	nickel 28 Ni 58.693	copper 29 Cu 63.546	zinc 30 Zn 65.38	gallium 31 Ga 69.723	germanium 32 Ge 72.64	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.798
rubidium 37 Rb 85.468	strontium 38 Sr 87.62	yttrium 39 Y 88.906	zirconium 40 Zr 91.224	niobium 41 Nb 92.906	molybdenum 42 Mo 95.96	technetium 43 Tc [98]	ruthenium 44 Ru 101.07	rhodium 45 Rh 102.91	palladium 46 Pd 106.42	silver 47 Ag 107.87	cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29
caesium 55 Cs 132.91	barium 56 Ba 137.33		hafnium 72 Hf 178.49	tantalum 73 Ta 180.95	tungsten 74 W 183.84	rhenium 75 Re 186.21	osmium 76 Os 190.23	iridium 77 Ir 192.22	platinum 78 Pt 195.08	gold 79 Au 196.97	mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]
francium 87 Fr [223]	radium 88 Ra [226]		rutherfordium 104 Rf [261]	dubnium 105 Db [262]	seaborgium 106 Sg [266]	bohrium 107 Bh [264]	hassium 108 Hs [277]	meitnerium 109 Mt [268]	darmstadtium 110 Ds [271]	roentgenium 111 Rg [272]							

lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.05	lutetium 71 Lu 174.97
actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]	lawrencium 103 Lr [262]

ICP-OES – summary

Pros	Cons
Tool of choice for analysis of common elements in geochemical fluids. Can measure 7/8 most abundant elements in crust & most trace elements.	Cannot analyse important geochemical species such as halogens (F, Cl, Br, I), CHON elements, and noble gases.
Rapid, cost effective, minimal sample preparation required.	Must account for matrix effects.
With acid digestion or laser ablation, can be used for solids analysis.	Spectral interference is main problem; need appropriate standards & calibrations.

What is ICP-MS?

Inductively **C**oupled **P**lasma – **M**ass **S**pectrometry

<u>Why using ICP-MS</u>?

1) Almost no bonds survive plasma heat, no chemical interferences (*well…*).
2) Background is low, so low detection limits.
3) Simple linear calibration over several orders of magnitude.

What is ICP-MS?

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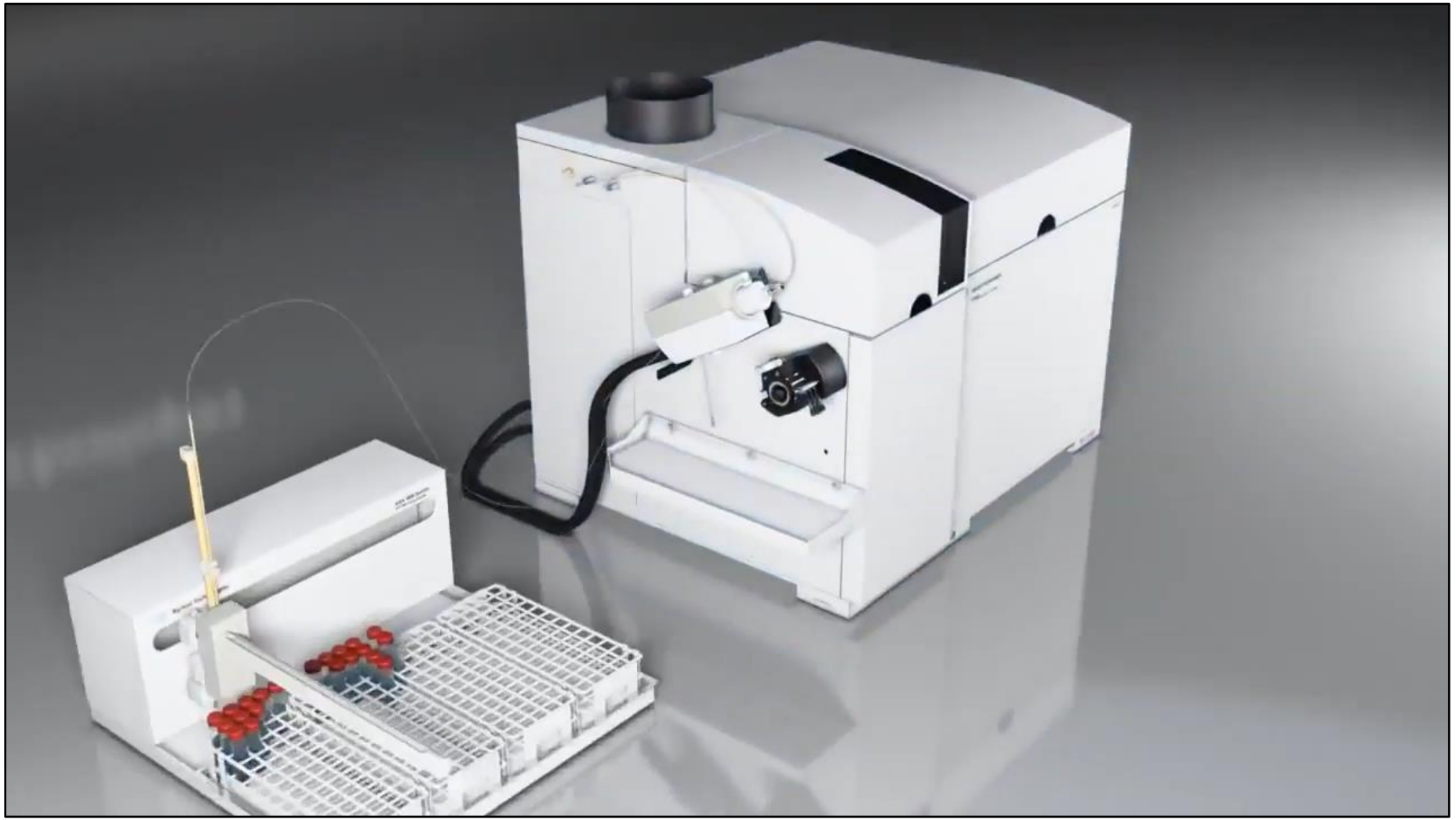
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<u>Key applications</u>

- Toxicology of metals hazardous at very low concentrations (As).
- Ultra-dilute fluid tracers (Cd, REE); groundwater paths, ocean circulation.
- Platinum-group elements – local redox fronts.

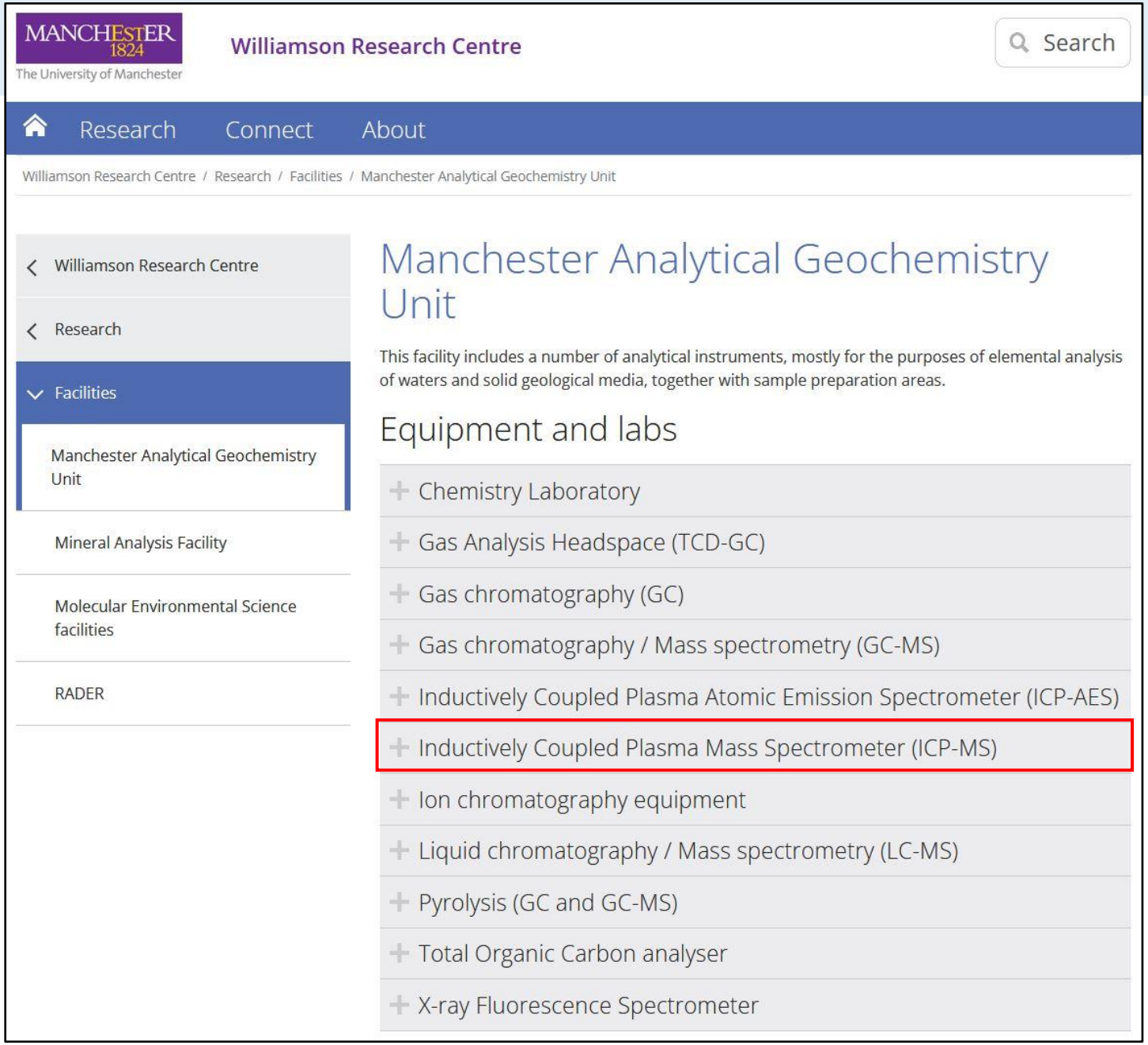
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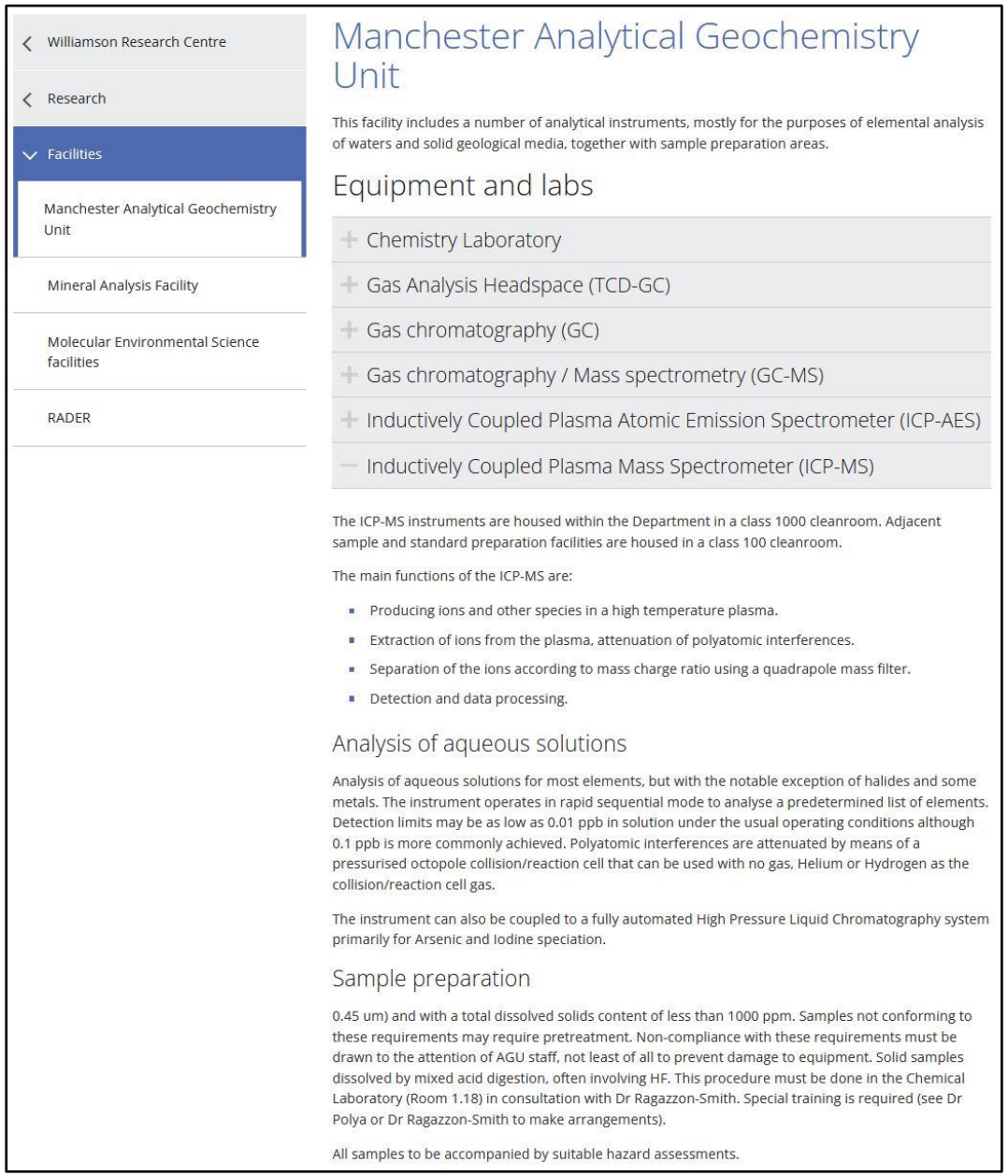
https://www.youtube.com/watch?v=rH_TEDa1qV0

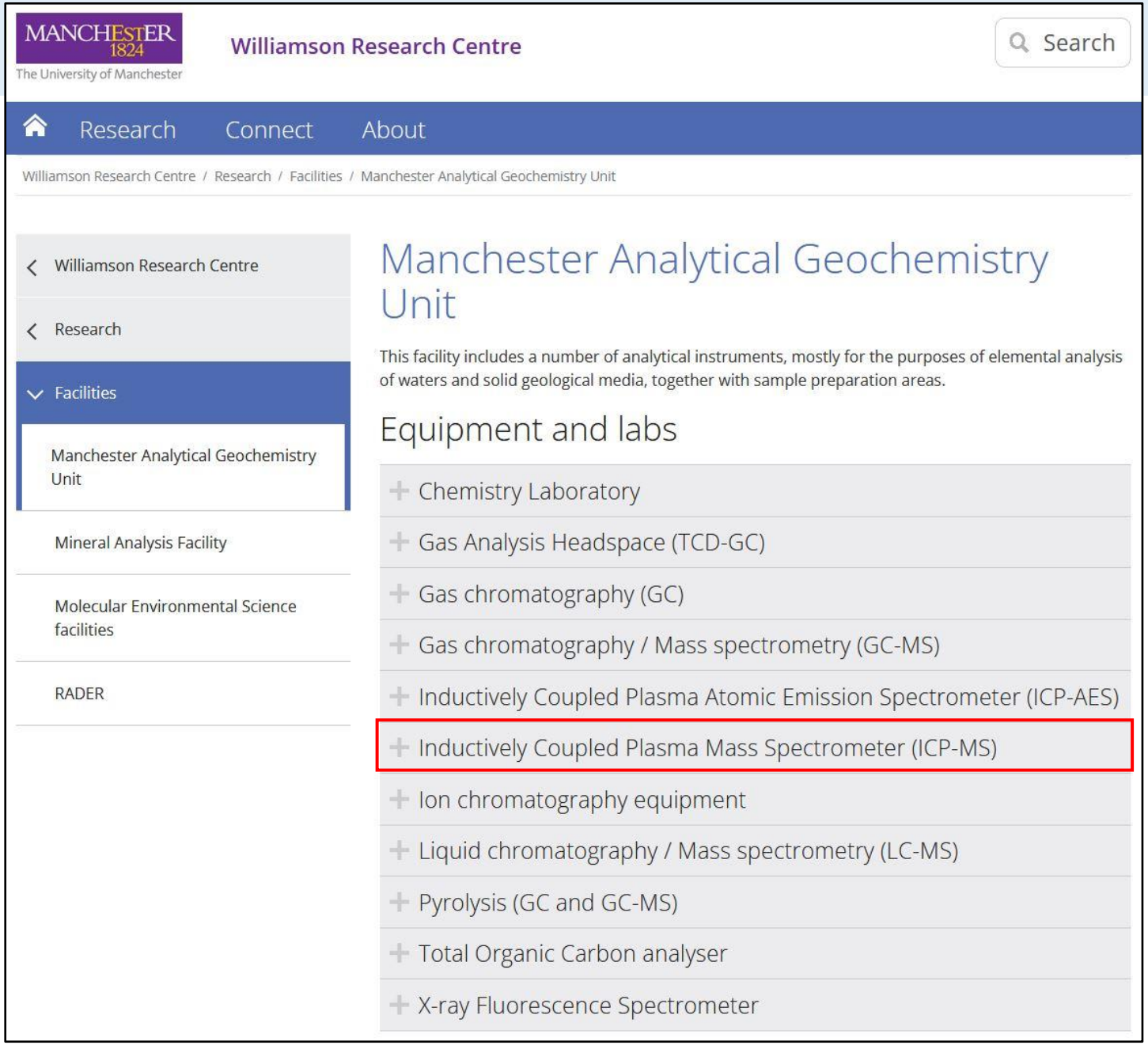
ICP-MS in Manchester



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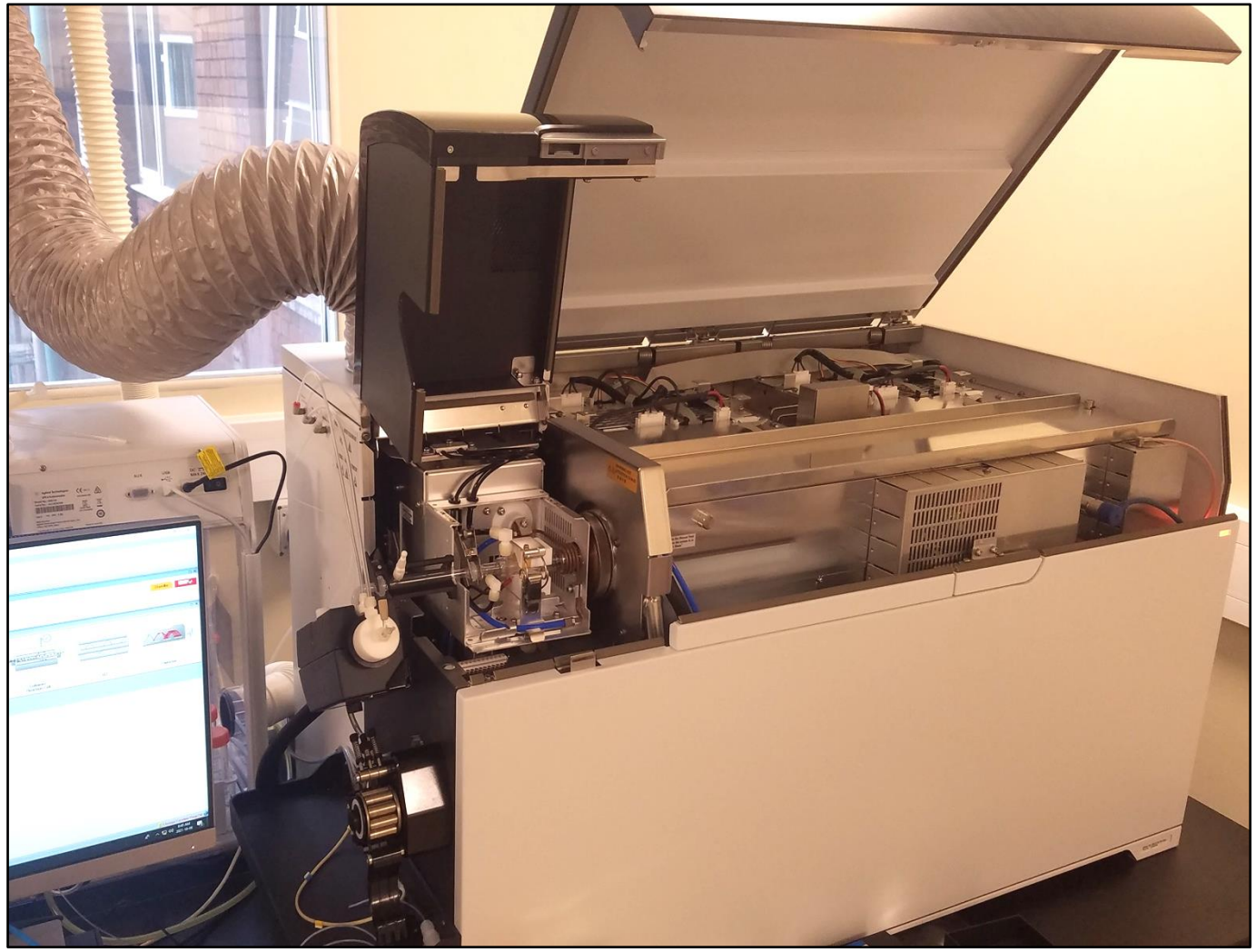
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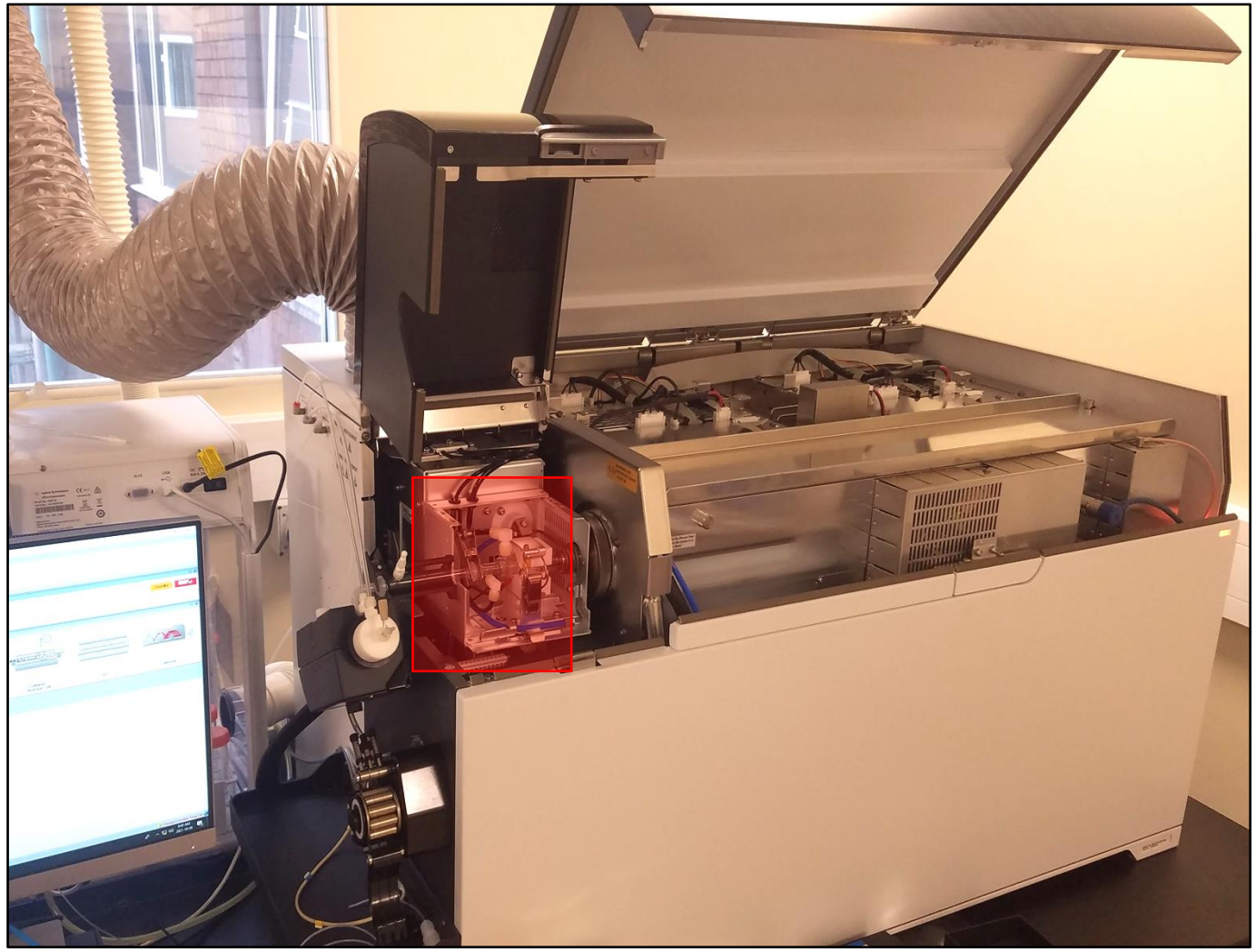
ICP-MS in Manchester



<u>Key components</u>:

1) Source with plasma torch: *ionise elements*
2) Interface with cones & electrostatic lens assembly: *skims ions from plasma and focuses beam*
3) Quadrupole mass filter: *select masses sequentially*
4) Electron multiplier: *measures species intensity*
5) Computer: *record data*

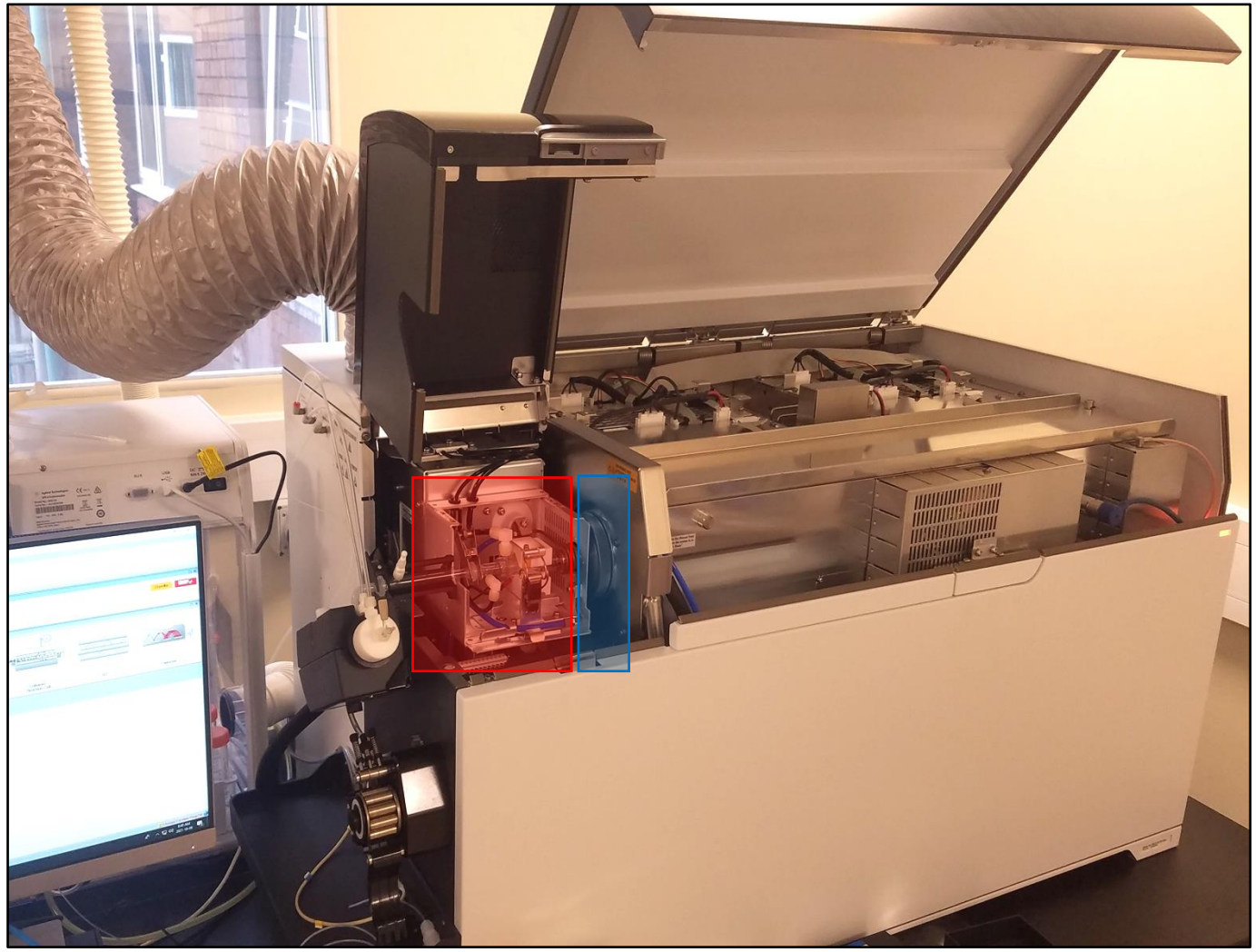
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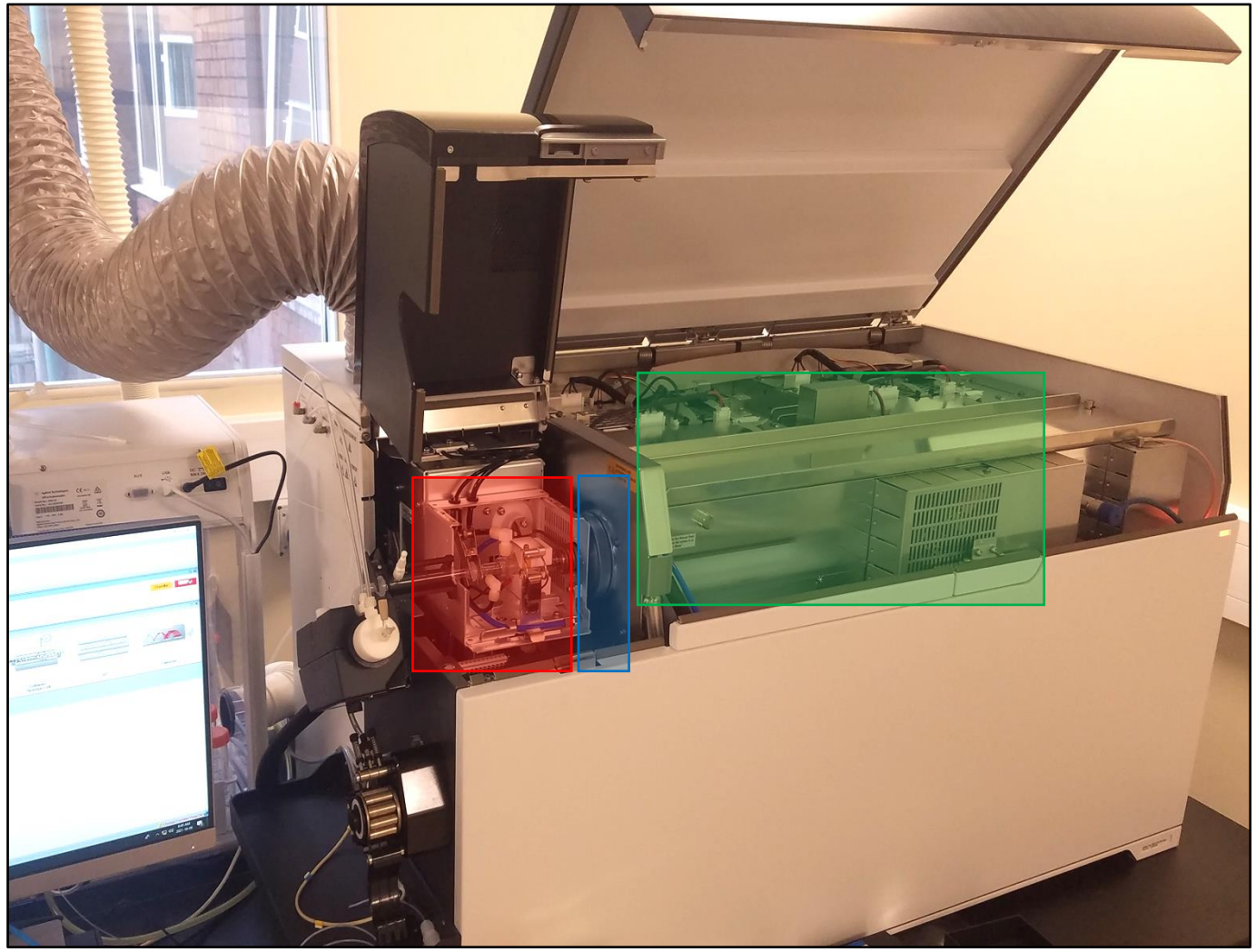
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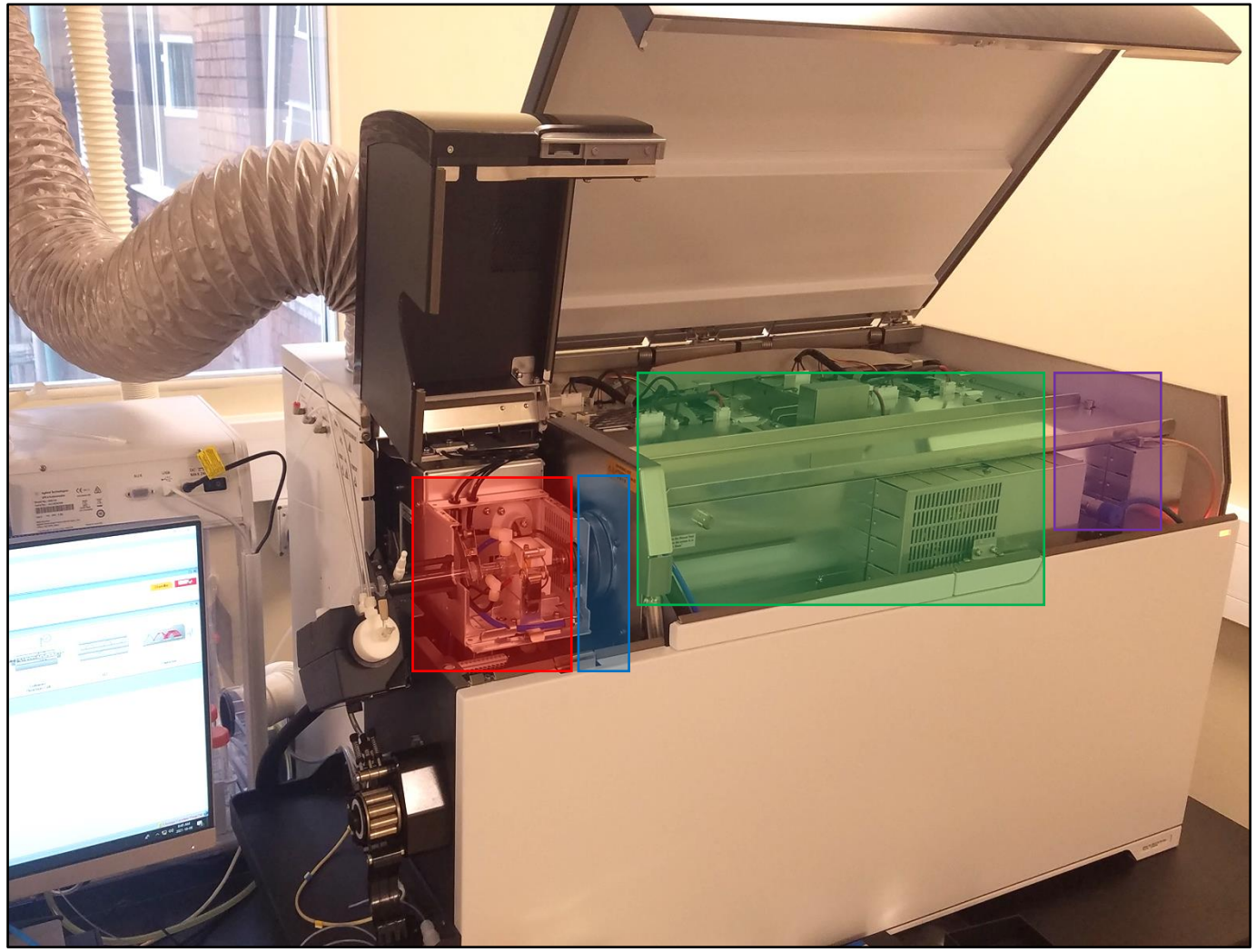
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Key components:

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2) Interface with cones & electrostatic lens assembly: *skims ions from plasma and focuses beam*
3) Quadrupole mass filter: *select masses sequentially*
4) Electron multiplier: *measures species intensity*
5) Computer: *record data*

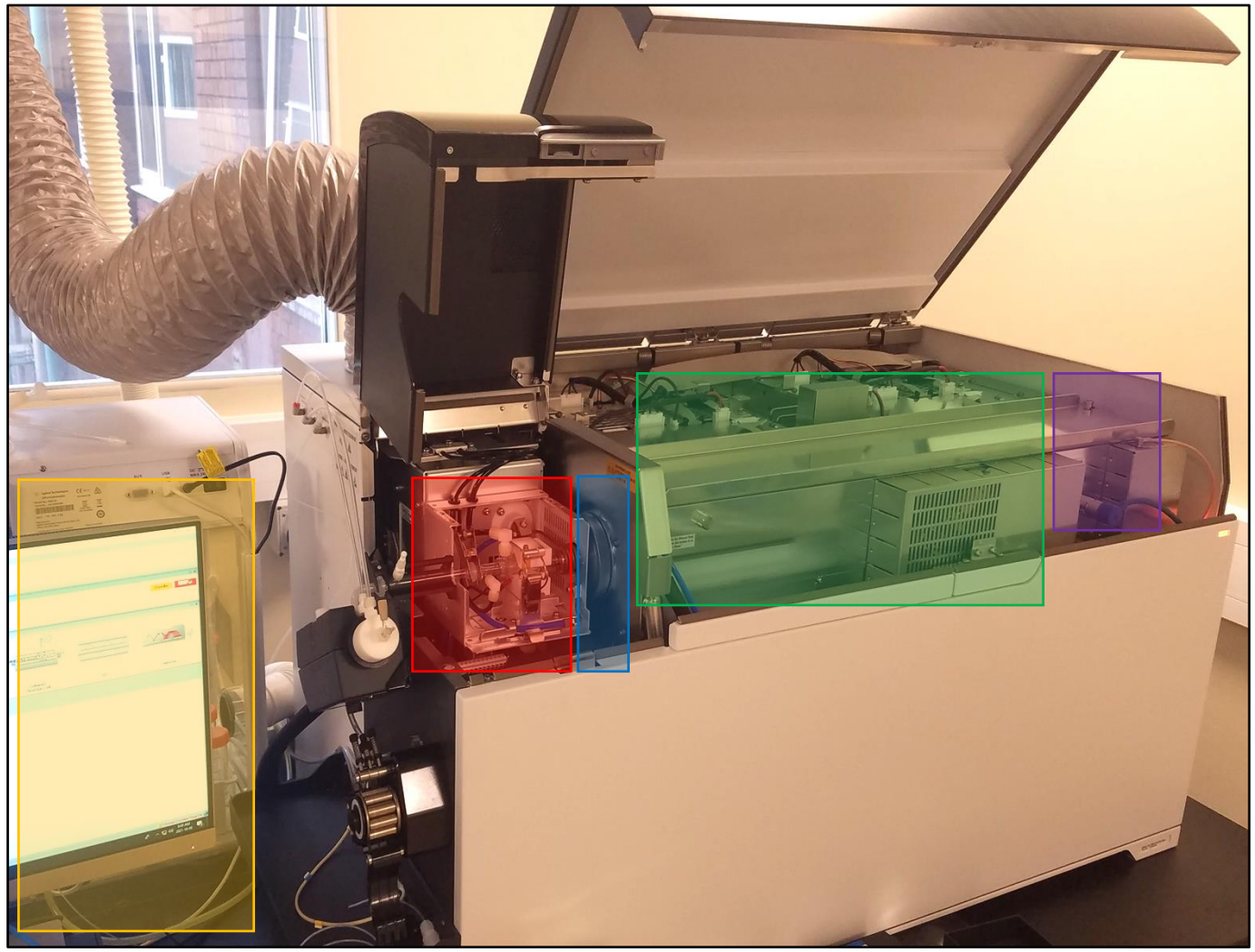
ICP-MS in Manchester



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ICP-MS in Manchester

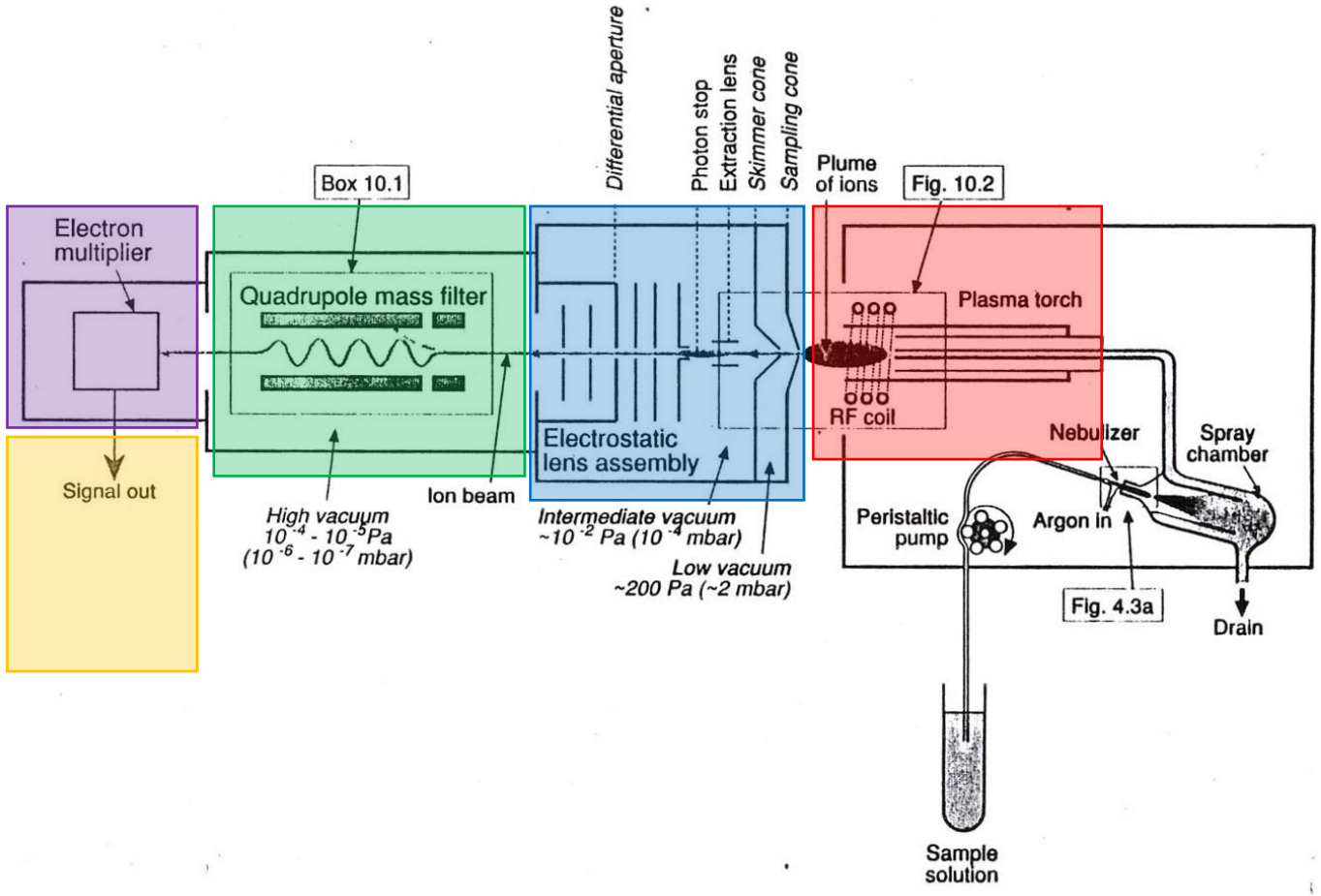
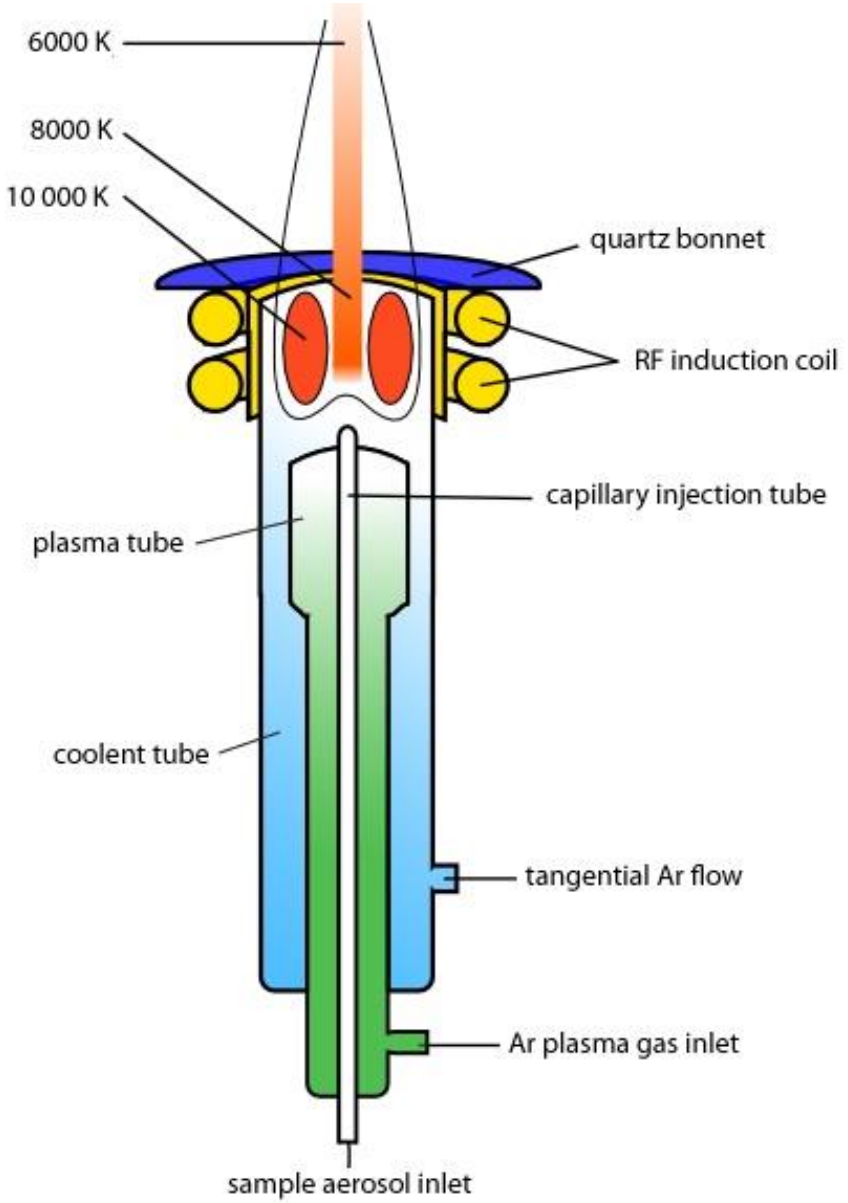


Figure 10.1 Schematic of an inductively coupled plasma–mass spectrometry instrument.

Key components:

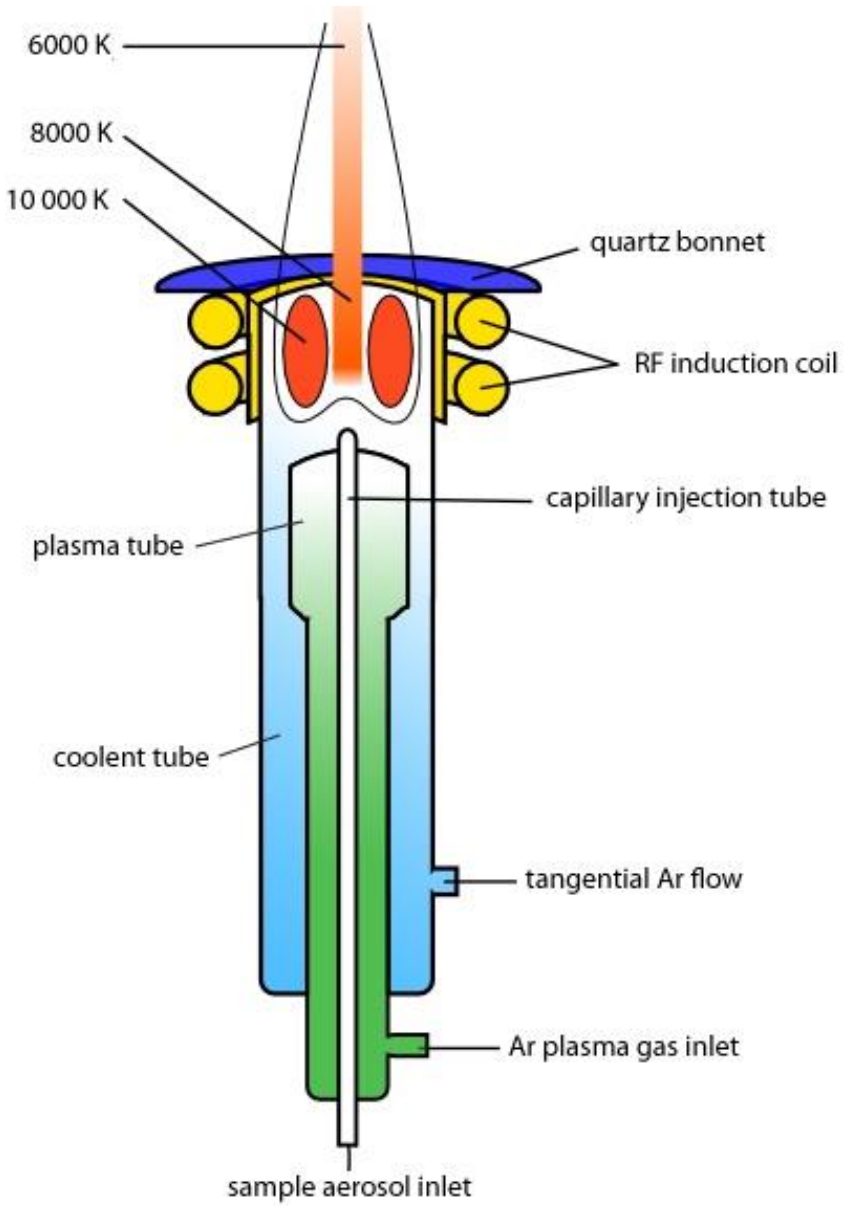
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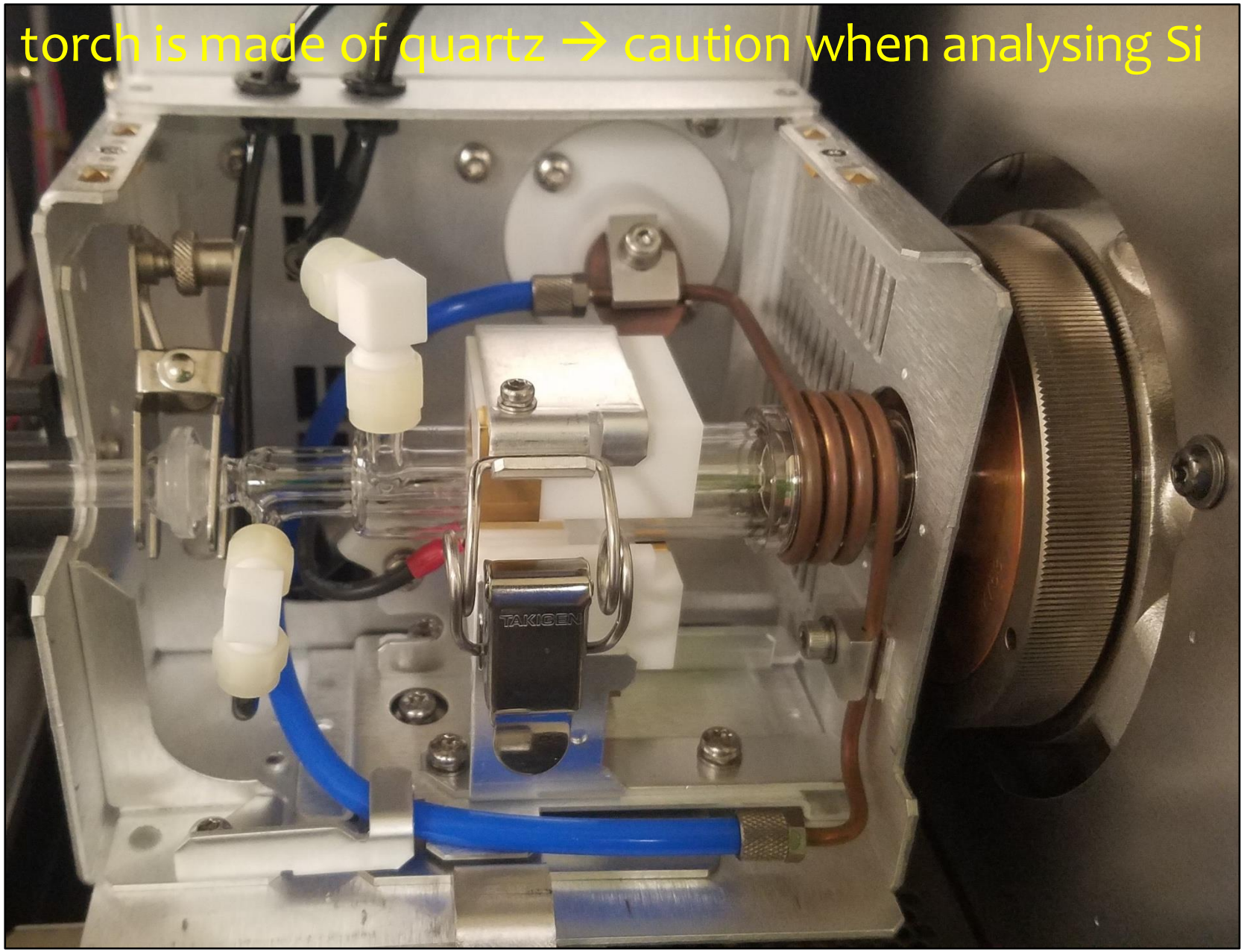
ICP-MS – torch and plasma



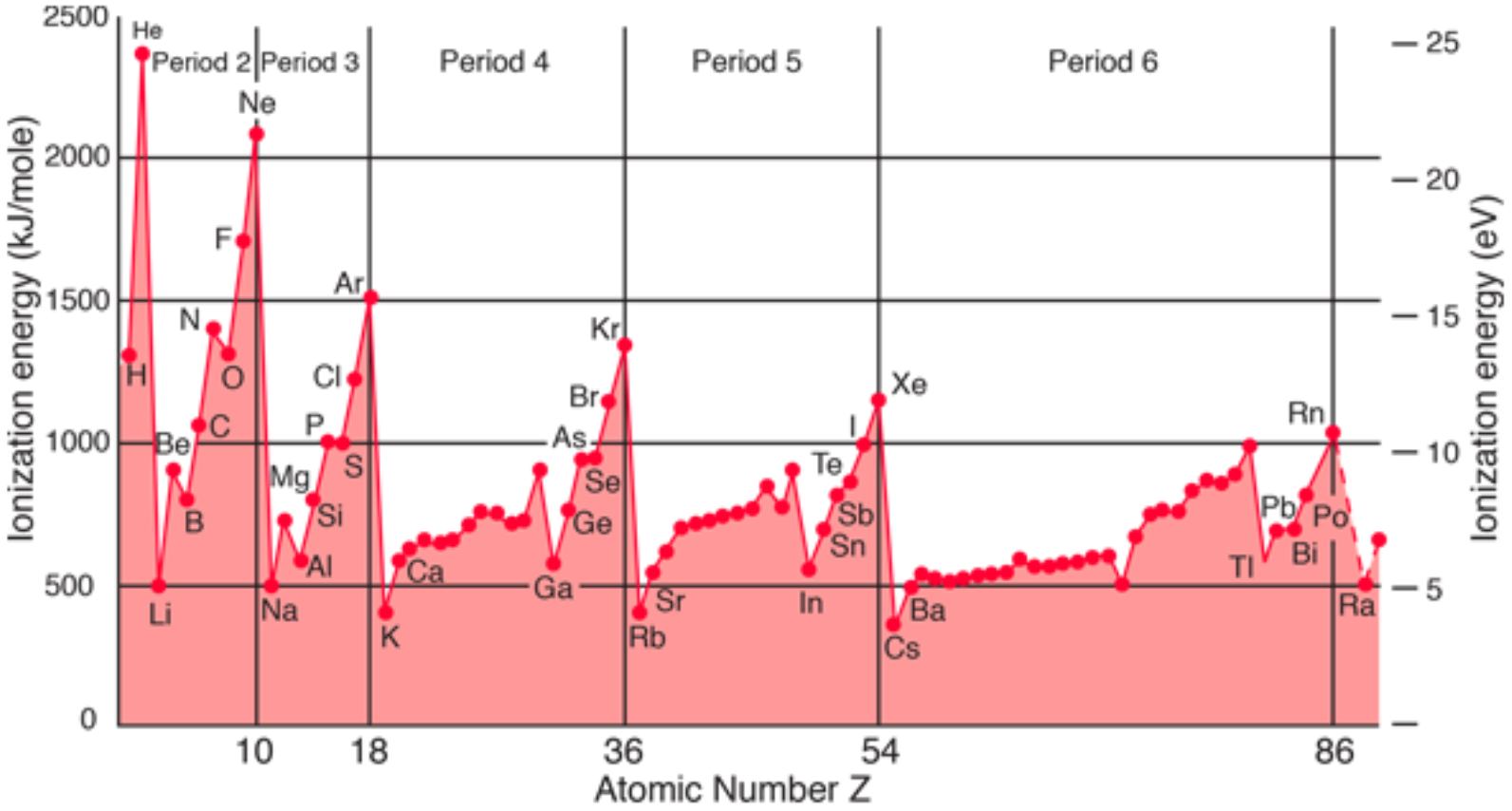
- Argon (Ar) gas flows in torch.
- Radio-frequency (RF) power applied to coil creates electric and magnetic fields at the end of the torch.
- Spark **strips electrons from Ar gas**; these electrons accelerate and increase energy in magnetic field; they **collide with Ar atoms** and strip off more electrons; eventually plasma contains Ar atoms, Ar^+ ions, and electrons.
- Introduced aerosol is (1) ***desolvated*** in solid molecules, which are then (2) ***vaporised*** in gas molecules, then (3) ***atomised*** into atoms.
- Finally, atoms are excited and/or ***ionised***.

ICP-MS – torch and plasma





ICP-MS – ionisation

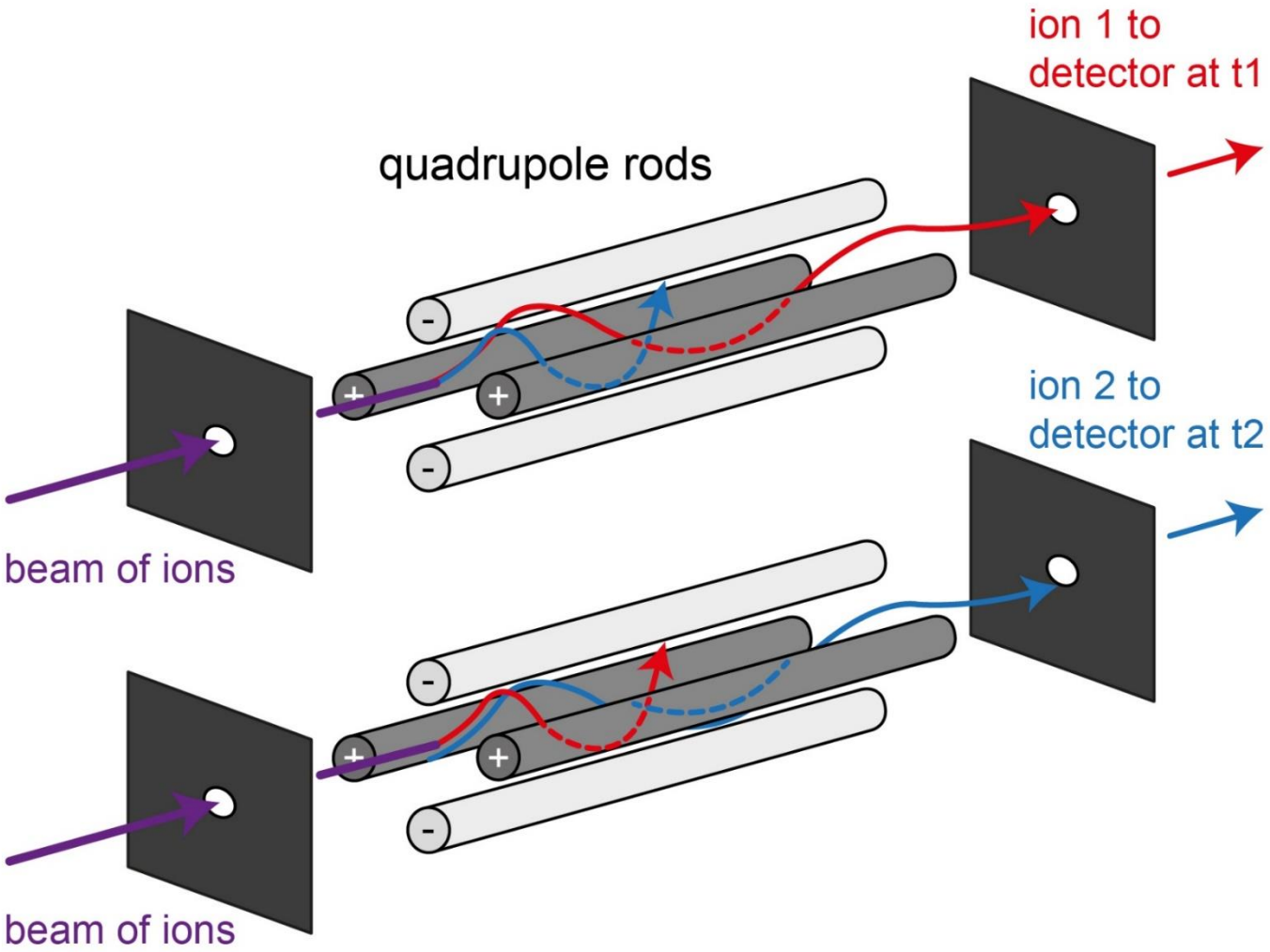


In theory, ICP-MS can analyse any element with ionisation potential below that of Ar.

- **Ionisation energy**, or potential, is the energy necessary to remove an electron from a neutral atom.
- It is lowest for the alkali metals, which have a single electron outside a closed shell.
- Generally increases across a row on the periodic table, max for noble gases, which have closed shells.

http://hyperphysics.phy-astr.gsu.edu/hbase/Chemical/ionize.html

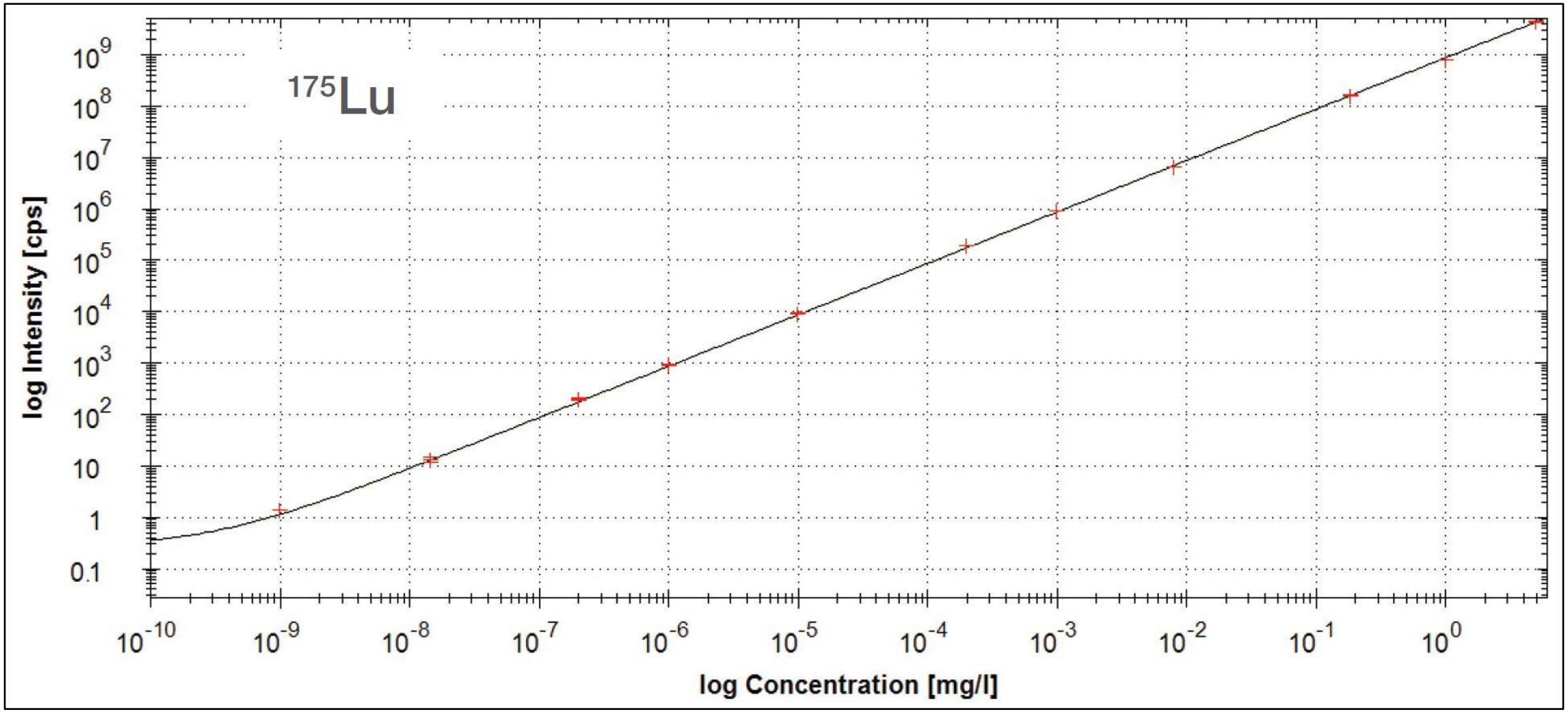
ICP-MS – mass filtering



- Applied potentials on opposed pairs of rods varies sinusoidally with time.
- Creates oscillations between the rods, resulting in complex ion trajectories dependent on their *m*/*z* (mass/charge ratio).
- Specific potential & frequency combinations result in specific ions being in resonance, creating a stable trajectory through the quadrupole to the detector for a defined *m*/*z* value.
- All other *m*/*z* values will be non-resonant and hit the rods.

ICP-MS – abundance quantification

- Linear dynamic range up to 10 orders of magnitude, **if matrix is constant**.



https://documents.thermofisher.com/TFS-Assets/CMD/Technical-Notes/tn-43399-icp-ms-linear-dynamic-range-tn43399-en.pdf

Elements measurable by ICP-MS

Detection limit range

- <0.1-1 ppt (ng/L)
- 1-10 ppt (ng/L)
- 10-100 ppt (ng/L)
- 0.1-1 ppb (µg/L)
- 1-10 ppb (µg/L)

"Identifying a 1 ppt element in a solution is analogous to locating a single white raisin in a 260 m^2 house full of regular raisins"

hydrogen 1 H 1.0079																	helium 2 He 4.0026
lithium 3 Li 6.941	beryllium 4 Be 9.0122											boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180
sodium 11 Na 22.990	magnesium 12 Mg 24.305											aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948
potassium 19 K 39.098	calcium 20 Ca 40.078	scandium 21 Sc 44.956	titanium 22 Ti 47.867	vanadium 23 V 50.942	chromium 24 Cr 51.996	manganese 25 Mn 54.938	iron 26 Fe 55.845	cobalt 27 Co 58.933	nickel 28 Ni 58.693	copper 29 Cu 63.546	zinc 30 Zn 65.38	gallium 31 Ga 69.723	germanium 32 Ge 72.64	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.798
rubidium 37 Rb 85.468	strontium 38 Sr 87.62	yttrium 39 Y 88.906	zirconium 40 Zr 91.224	niobium 41 Nb 92.906	molybdenum 42 Mo 95.96	technetium 43 Tc [98]	ruthenium 44 Ru 101.07	rhodium 45 Rh 102.91	palladium 46 Pd 106.42	silver 47 Ag 107.87	cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29
caesium 55 Cs 132.91	barium 56 Ba 137.33		hafnium 72 Hf 178.49	tantalum 73 Ta 180.95	tungsten 74 W 183.84	rhenium 75 Re 186.21	osmium 76 Os 190.23	iridium 77 Ir 192.22	platinum 78 Pt 195.08	gold 79 Au 196.97	mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]
francium 87 Fr [223]	radium 88 Ra [226]		rutherfordium 104 Rf [261]	dubnium 105 Db [262]	seaborgium 106 Sg [266]	bohrium 107 Bh [264]	hassium 108 Hs [277]	meitnerium 109 Mt [268]	darmstadtium 110 Ds [271]	roentgenium 111 Rg [272]							

lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.05	lutetium 71 Lu 174.97
actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]	lawrencium 103 Lr [262]

ICP-MS – summary

Pros	Cons
Tool of choice for analysis of low concentration trace elements in geochemical fluids. Can measure almost all periodic table with detection limit below 0.1 ppb.	Cannot analyse important geochemical species such as CHON elements, and noble gases.
Rapid, relatively cost effective, minimal sample preparation required.	Must account for matrix effects. Cannot analyse samples with high total dissolved solid abundances.
With acid digestion or laser ablation, can be used for solids analysis.	Mass interferences (isobaric, polyatomic ions $^{40}Ar^{35}Cl^{+}$ and $^{75}As^{+}$, doubly charged $^{88}Sr^{++}$ and $^{44}Ca^{+}$); need appropriate standards & calibrations.

ICP-OES vs. ICP-MS

ICP-OES	ICP-MS
Typical detection limit in the ppb range.	Typical detection limit in the ppt range.
Can analyse samples with high total dissolved solids (up to 30%).	Can only analyse samples with low total dissolved solids (up to 0.2%).
Method development fairly easy.	Method development more complex.
Instrument fairly cheap, lower running cost; about £2/sample in EES.	Costly instrument, higher running cost; about £6/sample in EES.